Propensity Score Analysis in R - Advanced

Statistical Horizons

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Section 3.8: Computer Lab: Running Propensity Score Subclassification in Conjunction with Cox Regression with R

This lab illustrates propensity score subclassification followed by a multivariate outcome analysis using a Cox proportional hazards model. We will also show an example of marginal mean weighting through stratification (MMWS), which is an alternative method of estimating marginal effects after stratum weighting (as opposed to aggregating the results).

Conceptual Review

The idea of subclassification was developed even before the development of propensity score analysis. Suppose x is a variable known to cause selection bias. Researchers can stratify the entire sample by subclasses of x such that within a stratum, study participants are homogeneous on x, and therefore, the estimation of treatment effect using the standard estimator is net of the influence of x, and the selection bias due to x is removed. However, this type of **exact subclassification** suffers from the same dimensionality problem as matching. When the number of covariates increases, the number of subclasses grows exponentially. Therefore, it is advantageous to use a single propensity score rather than several observed covariates to stratify, reducing the dimensionality in subclassification.

Propensity score subclassification has five intuitive steps: (1) Sort the sample by estimated propensity scores in an ascending order; (2) Divide the sample into K strata using quantiles (quintiles, deciles, etc.); (3) Evaluate the treatment effect by calculating mean difference of outcome and variance of difference between treated and control participants within each stratum, or by running a multivariate analysis of outcomes within each stratum as one does for samples generated by a randomized experiment; (4) Estimate the mean difference (ATE) for the whole sample (i.e., all K strata combined) through aggregation; and (5) Test whether the sample difference on outcome is statistically significant.[1]

Assumptions

- Sufficient overlap in the covariate distributions in the treated and control subpopulations (overlap assumption)
- Cut points of the subclasses are fixed
- Responses are independent across all of the subclasses
- Within each subclass, the responses from the treated and controls are independent

Strengths

- Allows analysis of an outcome variable that is virtually any type
- Accommodates most multivariate models, e.g., OLS, SEM, survival analysis, multinomial logit model, ordered logistic regression
- Retains a large size of the original sample
- Both ATE and ATT can be estimated

Limitations

- Sensitive to the violation of the overlap assumption (a potential solution is **trimming** or discarding observations with propensity scores outside a particular interval)

Load Packages

The `haven` and `sjlabelled` packages are used to load and clean Stata data files (.dta); the `MatchIt` package contains a function for subclassification; and the `cobalt` package contains functions for checking balance.

```
if (!require("pacman")) install.packages("pacman")
pacman::p_load(
  haven, sjlabelled, cobalt, MatchIt, tidyverse, broom, survival, PSweight, sandwich,
```

[1] Guo, S., & Fraser, M. W. (2010). *Propensity score analysis: Statistical methods and applications*. Sage Publications.

```
  kableExtra, here
)
```

Import Data

This dataset contains 2,758 children from the panel data of the National Survey of Child and Adolescent Well-Being (NSCAW). After a listwise deletion of missing data, the study sample comprises 2,723 children.[2]

The research question is whether caregivers' use of substance abuse services (`aodserv`) is predictive of the timing (`durm`) of a maltreatment re-report (`rrpt`) 18 months after the baseline interview. Of the 2,723 children included in the study, 294 (10.8%) had female caregivers who received substance abuse treatment, and the remaining 2,429 (89.2%) had female caregivers who did not receive treatment services. Various demographic and risk variables are used as covariates.

```
# Import data and remove Stata formats/labels
df <- read_dta("data/adv/sur_subclass.dta") %>%
  haven::zap_formats() %>%
  sjlabelled::remove_all_labels() %>%
  as_tibble()

# Inspect data and treatment condition
head(df)
```

```
## # A tibble: 6 x 40
##    durm  rrpt PSH17A CRA47A aodserv married  high bahigh poverty2 poverty3
##   <dbl> <dbl>  <dbl>  <dbl>   <dbl>   <dbl> <dbl>  <dbl>    <dbl>    <dbl>
## 1    18     0      1      0       1       1     0      1        0        1
## 2    18     0      0      0       0       0     1      0        0        0
## 3    18     0      0      0       0       1     1      0        0        0
## 4    18     0      0      0       1       0     0      0        1        0
## 5    18     0      0      1       0       0     0      0        1        0
## 6    18     0      0      1       0       0     1      0        1        0
## # i 30 more variables: poverty4 <dbl>, poverty5 <dbl>, employ <dbl>,
## #   open <dbl>, black <dbl>, hispanic <dbl>, natam <dbl>, cgrage1 <dbl>,
## #   cgrage2 <dbl>, cgrage3 <dbl>, chdage1 <dbl>, chdage2 <dbl>, chdage3 <dbl>,
## #   mental <dbl>, arrest <dbl>, sexual <dbl>, provide <dbl>, supervis <dbl>,
## #   other <dbl>, ra <dbl>, cidi <dbl>, cgneed <dbl>, cwwrep <dbl>, id <dbl>,
## #   educ <dbl>, pov <dbl>, race <dbl>, chdage <dbl>, cgage <dbl>, maltx <dbl>
```

```
dim(df)
```

```
## [1] 2723   40
```

```
table(df$aodserv)
```

```
##
##    0    1
## 2429  294
```

Create Propensity Scores Using Logistic Regression

Here we calculate the probability of treatment assignment conditional on the known covariates using logistic regression. We obtain predicted probabilities from the logistic regression model as the propensity scores and add them to our orginal dataframe.

[2] Guo, S., Barth, R. P., & Gibbons, C. (2006). Propensity score matching strategies for evaluating substance abuse services for child welfare clients. *Children and Youth Services Review*, *28*(4), 357–383. https://doi.org/10.1016/j.childyouth.2005.04.012

```r
# Model specification for propensity score
(f <- cobalt::f.build("aodserv", select(df, married:cgneed, PSH17A:CRA47A)))
```

```
## aodserv ~ married + high + bahigh + poverty2 + poverty3 + poverty4 +
##     poverty5 + employ + open + black + hispanic + natam + cgrage1 +
##     cgrage2 + cgrage3 + chdage1 + chdage2 + chdage3 + mental +
##     arrest + sexual + provide + supervis + other + ra + cidi +
##     cgneed + PSH17A + CRA47A
## <environment: 0x00000123ee214f20>
```

```r
# Logistic regression
m1 <- glm(f, family = binomial, data = df)

# Obtain predicted probabilities as propensity scores
ps <- predict(m1, type = "response")

# Append predicted probabilities to original dataframe
df.ps <- df %>%
  mutate(ps = ps)
```

Check Balance Before Subclassification

Function to Check Balance Using Normalized Difference

```r
nd <- function(cov, data = df.ps, tx = "aodserv") {
  tx <- as.name(tx)
  cov <- as.name(cov)
  data <- data %>%
    group_by(!!tx) %>%
    summarize(
      m = mean(!!cov, na.rm = T),
      s = sd(!!cov, na.rm = T),
      .groups = "drop"
    ) %>%
    pivot_wider(names_from = !!tx, values_from = c("m", "s"), names_sep = "") %>%
    mutate(nd = (m1 - m0) / sqrt(s1^2 + s0^2))
  return(data$nd)
}
```

Check Balance Using Normalized Differences

Using the normalized difference (Imbens & Wooldridge, 2009), the following covariates have a difference that exceeds .25, indicating that selection bias exists: case status open (`open`), child age group of 0 to 2 years old (`chdage1`), caregiver mental health problem (`mental`), caregiver arrested (`arrest`), caregiver received alcohol or drug (AOD) treatment (`PSH17A`), presence of risk (`ra`), presence of the Composite International Diagnostic Interview–Short Form (CIDI-SF) (`cidi`), and caregiver report of need (`cgneed`).

```r
# Get a Vector of Covariates
(covs <- select(df, married:cgneed, PSH17A:CRA47A) %>% names())
```

```
##  [1] "married"  "high"     "bahigh"   "poverty2" "poverty3" "poverty4"
##  [7] "poverty5" "employ"   "open"     "black"    "hispanic" "natam"
## [13] "cgrage1"  "cgrage2"  "cgrage3"  "chdage1"  "chdage2"  "chdage3"
## [19] "mental"   "arrest"   "sexual"   "provide"  "supervis" "other"
## [25] "ra"       "cidi"     "cgneed"   "PSH17A"   "CRA47A"
```

```r
# Apply Function "nd" on Each Covariate to get Normalized Differences
sapply(covs, nd) %>% round(., 3) %>% sort()
```

```
##   sexual   employ  chdage3   bahigh poverty3  married  chdage2 poverty4
##   -0.236   -0.206   -0.168   -0.117   -0.102   -0.084   -0.061   -0.059
##  cgrage1 poverty5 hispanic     high    other  cgrage3  cgrage2 supervis
##   -0.059   -0.031   -0.031   -0.021    0.009    0.027    0.028    0.043
## poverty2    black    natam   CRA47A  provide  chdage1   cgneed     open
##    0.048    0.057    0.113    0.178    0.221    0.317    0.355    0.366
##   mental   PSH17A   arrest     cidi       ra
##    0.400    0.458    0.492    0.530    0.996
```

```r
# Select Normalized Differences Greater than the .25 Threshold
sapply(covs, nd) %>% .[abs(.) > .25] %>% round(., 3) %>% sort()
```

```
## chdage1  cgneed    open  mental  PSH17A  arrest    cidi      ra
##   0.317   0.355   0.366   0.400   0.458   0.492   0.530   0.996
```

Check Balance Using Standardized Mean Differences

Several R packages (e.g., `MatchIt` and `cobalt`) that contain functions for balance checking use a similar statistic, the standardized mean difference (SMD). In the cobalt package, the denominator in the SMD is by default the square root of the average of the variances of a variable in the treated and control groups (while the normalized difference uses the square root of the sums of the sample variances). This use of the pooled standard deviation in the SMD formula is most appropriate when estimating the ATE. When the ATT is desired, the standard deviation of the treated group is most appropriate. Sampling weights can be optionally included in the `bal.tab` function via the `s.weights` argument.[3]

When using a criteria of 0.1 for the SMD, 17 variables do not have satisfactory balance, as seen below:

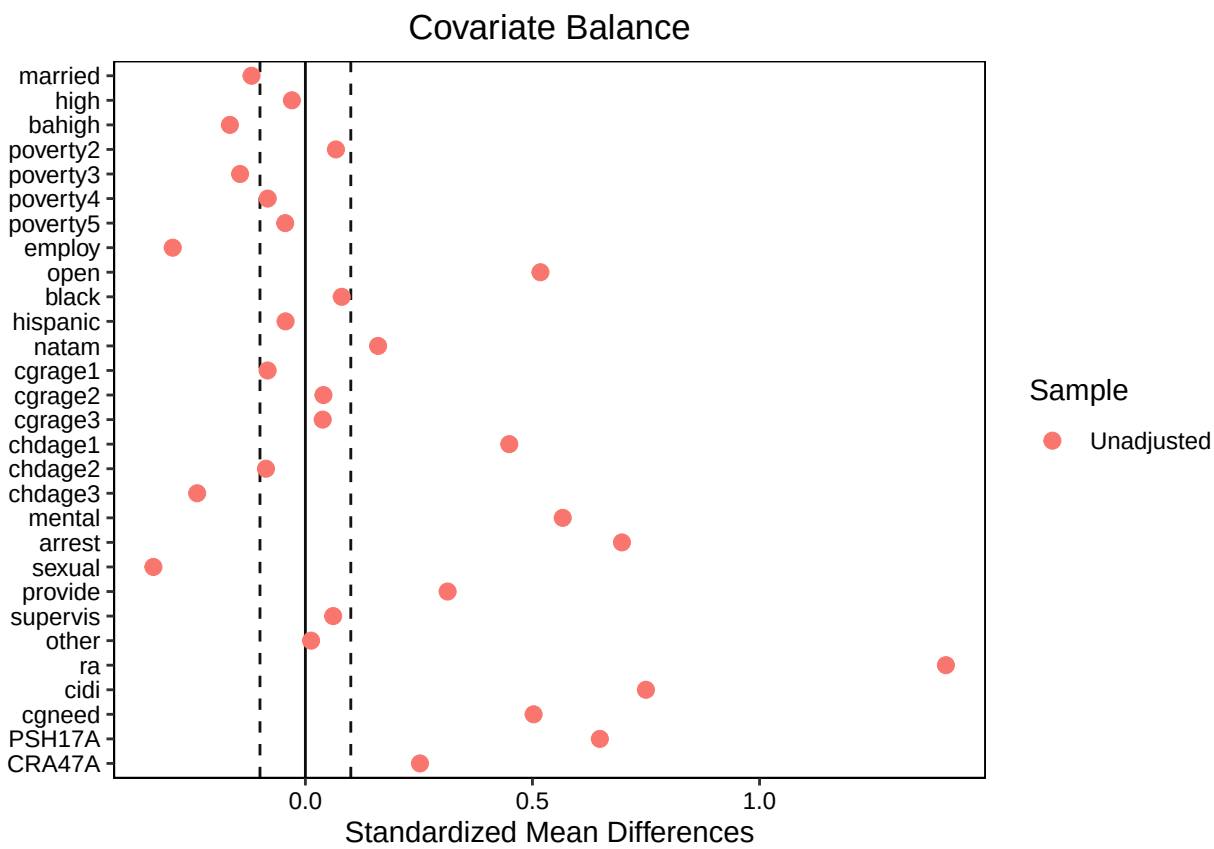
```r
# Standardized Mean Differences with .1 Threshold
(smds <- cobalt::bal.tab(f, data = df.ps, threshold = c(m = .1), s.d.denom = "pooled",
                binary = "std", abs = F))
```

```
## Balance Measures
##            Type Diff.Un     M.Threshold.Un
## married  Binary -0.1186 Not Balanced, >0.1
## high     Binary -0.0296     Balanced, <0.1
## bahigh   Binary -0.1663 Not Balanced, >0.1
## poverty2 Binary  0.0673     Balanced, <0.1
## poverty3 Binary -0.1438 Not Balanced, >0.1
## poverty4 Binary -0.0829     Balanced, <0.1
## poverty5 Binary -0.0444     Balanced, <0.1
## employ   Binary -0.2922 Not Balanced, >0.1
## open     Binary  0.5176 Not Balanced, >0.1
## black    Binary  0.0800     Balanced, <0.1
## hispanic Binary -0.0436     Balanced, <0.1
## natam    Binary  0.1601 Not Balanced, >0.1
## cgrage1  Binary -0.0832     Balanced, <0.1
## cgrage2  Binary  0.0397     Balanced, <0.1
## cgrage3  Binary  0.0381     Balanced, <0.1
## chdage1  Binary  0.4490 Not Balanced, >0.1
## chdage2  Binary -0.0868     Balanced, <0.1
## chdage3  Binary -0.2383 Not Balanced, >0.1
## mental   Binary  0.5666 Not Balanced, >0.1
```

[3] https://ngreifer.github.io/cobalt/articles/cobalt.html

```
## arrest   Binary  0.6970 Not Balanced, >0.1
## sexual   Binary -0.3347 Not Balanced, >0.1
## provide  Binary  0.3132 Not Balanced, >0.1
## supervis Binary  0.0612     Balanced, <0.1
## other    Binary  0.0125     Balanced, <0.1
## ra       Binary  1.4104 Not Balanced, >0.1
## cidi     Binary  0.7498 Not Balanced, >0.1
## cgneed   Binary  0.5026 Not Balanced, >0.1
## PSH17A   Binary  0.6484 Not Balanced, >0.1
## CRA47A   Binary  0.2523 Not Balanced, >0.1
##
## Balance tally for mean differences
##                     count
## Balanced, <0.1         12
## Not Balanced, >0.1     17
##
## Variable with the greatest mean difference
##  Variable Diff.Un     M.Threshold.Un
##        ra  1.4104 Not Balanced, >0.1
##
## Sample sizes
##     Control Treated
## All    2429     294
```

```
cobalt::love.plot(smds)
```



Function to Check Balance Using Chi-Square Tests

Chi-square tests can also be used to assess balance. However, hypothesis tests are influenced by sample size, which can change due to adjustments such as sample trimming.

```
check_blc <- function(cov, df = df.ps, tx = "aodserv") {
  ct <- broom::tidy(
    chisq.test(
      table(df[[cov]], y = df[[tx]]),
      correct = F
    )
  ) %>%
    add_column(cov = cov, .before = "statistic") %>%
    mutate(p.value = round(p.value, 2))
  return(ct)
}
```

Check Balance Using Chi-Square Tests

```
covs2 <- c(
  "married", "educ", "pov", "employ", "open", "race", "chdage",
  "cgage", "CRA47A", "mental", "arrest", "PSH17A", "maltx", "ra",
  "cidi", "cgneed"
)
map_dfr(covs2, check_blc)
```

```
## Warning in chisq.test(table(df[[cov]], y = df[[tx]]), correct = F): Chi-squared
## approximation may be incorrect
```

```
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married      3.52    0.06         1 Pearson's Chi-squared test
##  2 educ        10.2     0.01         2 Pearson's Chi-squared test
##  3 pov         11.2     0.02         4 Pearson's Chi-squared test
##  4 employ      21.5     0            1 Pearson's Chi-squared test
##  5 open        60.6     0            1 Pearson's Chi-squared test
##  6 race        11.6     0.01         3 Pearson's Chi-squared test
##  7 chdage      56.0     0            3 Pearson's Chi-squared test
##  8 cgage        4.00    0.26         3 Pearson's Chi-squared test
##  9 CRA47A      17.7     0            1 Pearson's Chi-squared test
## 10 mental      92.4     0            1 Pearson's Chi-squared test
## 11 arrest     127.      0            1 Pearson's Chi-squared test
## 12 PSH17A     171.      0            1 Pearson's Chi-squared test
## 13 maltx       47.9     0            4 Pearson's Chi-squared test
## 14 ra         584.      0            1 Pearson's Chi-squared test
## 15 cidi       154.      0            1 Pearson's Chi-squared test
## 16 cgneed     134.      0            1 Pearson's Chi-squared test
```

```
# Use Fisher's exact test for cgage (because expected cell count < 5)
table(df.ps$cgage, df.ps$aodserv)
```

```
##
##        0    1
##   0   35    8
##   1 1689  193
```

```
##   2  570   74
##   3  135   19
```

```r
(cq <- chisq.test(df.ps$cgage, df.ps$aodserv))
```

```
## Warning in chisq.test(df.ps$cgage, df.ps$aodserv): Chi-squared approximation
## may be incorrect
```

```
##
##  Pearson's Chi-squared test
##
## data:  df.ps$cgage and df.ps$aodserv
## X-squared = 3.9969, df = 3, p-value = 0.2618
```

```r
cq$expected
```

```
##            df.ps$aodserv
## df.ps$cgage          0          1
##           0   38.35733   4.642674
##           1 1678.80206 203.197943
##           2  574.46787  69.532134
##           3  137.37275  16.627249
```

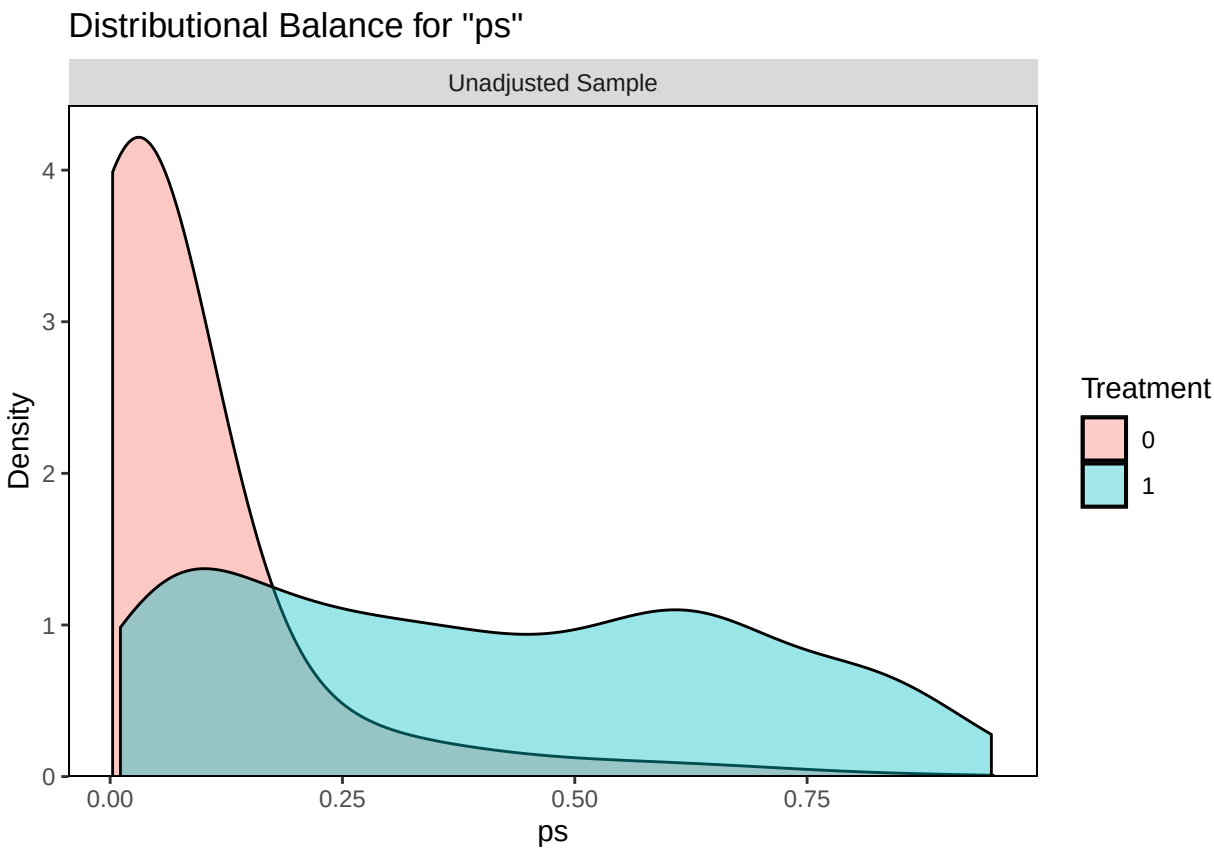
```r
fisher.test(df.ps$cgage, df.ps$aodserv)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  df.ps$cgage and df.ps$aodserv
## p-value = 0.2387
## alternative hypothesis: two.sided
```

Check Overlap Assumption

There is no significant overlap between the propensity score distributions of the treated and control groups.

```r
cobalt::bal.plot(x = df.ps, var.name = "ps", treat = df.ps$aodserv)
```



Create Subclasses

The crucial condition for a successful propensity score subclassification is to make the propensity score constant within a stratum. Previous studies have shown that a subclassification of five groups efficiently removes 90% of the bias (Cochran, 1968; Rosenbaum & Rubin, 1984). While using more subclasses creates better homogeneity within subclasses and reduces bias in treatment effect estimates, it also results in fewer observations in each subclass and therefore less precise treatment estimates (Du, 1998). Ultimately, we need to test different numbers of strata and choose one that optimizes the estimation of the treatment effect, while balancing covariates within each stratum.[4]

Get Quintiles

The function `quantile` produces sample quantiles corresponding to the given probabilities. The argument `type` specifies one of nine quantile algorithms. See `?quantile` for details.

```
# Produce quintiles and round to 7 digits to replicate the Stata results
(q5 <- c(0, round(quantile(df.ps$ps, c(.2, .4, .6, .8), type = 1), 7), 1))
```

```
##                 20%       40%       60%       80%
## 0.0000000 0.0121536 0.0216452 0.0440495 0.1421116 1.0000000
```

Create Five Subclasses Using Quintiles

Here we sort the propensity score in an ascending order and use the quintiles calculated above to divide the sample into five subclasses. See `?cut` for details on the various arguments used below. Note that we will preserve the grouping structure with the `dplyr::group_by()` function, which will allow us to carry out additional group-specific operations (e.g., using the function `dplyr::group_map()`).

[4] Guo, S., & Fraser, M. W. (2010). *Propensity score analysis: Statistical methods and applications*. Sage Publications.

```r
# Cut 5 subclasses
subs <- df.ps %>%
  arrange(ps) %>%
  group_by(q = cut(
    ps,
    include.lowest = F,
    right = F,
    dig.lab = 7,
    breaks = q5,
    labels = F
  ))

# Summary statistics per subclass
# Notice the severe violation of the overlap assumption
subs %>%
  group_by(q) %>%
  summarise(
    obs = n(),
    mean = mean(ps),
    sd = sd(ps),
    min = min(ps),
    max = max(ps), .groups = "drop"
  ) %>%
  left_join(
    subs %>%
      count(q, aodserv) %>%
      pivot_wider(names_from = "aodserv", values_from = "n", names_prefix = "aodserv"),
    by = "q"
  )
```

```
## # A tibble: 5 x 8
##       q   obs    mean      sd     min    max aodserv0 aodserv1
##   <int> <int>   <dbl>   <dbl>   <dbl>  <dbl>    <int>    <int>
## 1     1   545 0.00813 0.00237 0.00263 0.0122      544        1
## 2     2   545 0.0165  0.00271 0.0122  0.0216      539        6
## 3     3   543 0.0308  0.00600 0.0217  0.0439      529       14
## 4     4   546 0.0758  0.0264  0.0440  0.142       496       50
## 5     5   544 0.409   0.215   0.143   0.951       321      223
```

Check Balance After Subclassification

It is important to check balance on the observed covariates between the two groups in each stratum before proceeding to the next step. The results are not positive. As is often the case, the two groups did not balance on certain covariates, particularly at the two ends of the five subclasses. Note that all participants in Subclass 1 had low propensity scores and that all participants in Subclass 5 had high scores. As such, some covariates show statistical significant differences between the two treatment groups—that is, they were not balanced between treatment conditions. Alarmingly, in the highest and lowest subclasses, some participants fell uniquely into one category of certain covariates.

```r
# Check balance on each group in subs
subs %>%
  group_map(~ map_dfr(covs2, check_blc, df = .x))
```

```
## [[1]]
## # A tibble: 16 x 5
```

```
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <dbl> <chr>
##  1 married   0.506      0.48         1 Pearson's Chi-squared test
##  2 educ      0.955      0.62         2 Pearson's Chi-squared test
##  3 pov       2.35       0.67         4 Pearson's Chi-squared test
##  4 employ    0.478      0.49         1 Pearson's Chi-squared test
##  5 open      2.69       0.1          1 Pearson's Chi-squared test
##  6 race      0.673      0.88         3 Pearson's Chi-squared test
##  7 chdage    2.31       0.51         3 Pearson's Chi-squared test
##  8 cgage     3.59       0.17         2 Pearson's Chi-squared test
##  9 CRA47A    0.211      0.65         1 Pearson's Chi-squared test
## 10 mental    0.0303     0.86         1 Pearson's Chi-squared test
## 11 arrest    0.0402     0.84         1 Pearson's Chi-squared test
## 12 PSH17A    0.00741    0.93         1 Pearson's Chi-squared test
## 13 maltx     2.39       0.66         4 Pearson's Chi-squared test
## 14 ra      541.         0            1 Chi-squared test for given probabilities
## 15 cidi      0.0323     0.86         1 Pearson's Chi-squared test
## 16 cgneed  541.         0            1 Chi-squared test for given probabilities
##
## [[2]]
## # A tibble: 16 x 5
##    cov      statistic p.value parameter method
##    <chr>        <dbl>   <dbl>     <dbl> <chr>
##  1 married   0.749       0.39         1 Pearson's Chi-squared test
##  2 educ      2.12        0.35         2 Pearson's Chi-squared test
##  3 pov       1.77        0.78         4 Pearson's Chi-squared test
##  4 employ    3.02        0.08         1 Pearson's Chi-squared test
##  5 open      2.21        0.14         1 Pearson's Chi-squared test
##  6 race      4.68        0.2          3 Pearson's Chi-squared test
##  7 chdage    5.67        0.13         3 Pearson's Chi-squared test
##  8 cgage     0.832       0.84         3 Pearson's Chi-squared test
##  9 CRA47A    1.81        0.18         1 Pearson's Chi-squared test
## 10 mental    1.03        0.31         1 Pearson's Chi-squared test
## 11 arrest    0.000198    0.99         1 Pearson's Chi-squared test
## 12 PSH17A    0.0789      0.78         1 Pearson's Chi-squared test
## 13 maltx     2.24        0.69         4 Pearson's Chi-squared test
## 14 ra      521.          0            1 Chi-squared test for given probabilities
## 15 cidi      0.250       0.62         1 Pearson's Chi-squared test
## 16 cgneed   44.1         0            1 Pearson's Chi-squared test
##
## [[3]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married   0.147      0.7          1 Pearson's Chi-squared test
##  2 educ      2.85       0.24         2 Pearson's Chi-squared test
##  3 pov       1.41       0.84         4 Pearson's Chi-squared test
##  4 employ    1.93       0.17         1 Pearson's Chi-squared test
##  5 open      0.260      0.61         1 Pearson's Chi-squared test
##  6 race      3.90       0.27         3 Pearson's Chi-squared test
##  7 chdage    1.07       0.78         3 Pearson's Chi-squared test
##  8 cgage     3.15       0.37         3 Pearson's Chi-squared test
##  9 CRA47A    0.428      0.51         1 Pearson's Chi-squared test
## 10 mental    0.447      0.5          1 Pearson's Chi-squared test
```

```
## 11 arrest    0.00553    0.94         1 Pearson's Chi-squared test
## 12 PSH17A   10.5        0            1 Pearson's Chi-squared test
## 13 maltx     3.94       0.41         4 Pearson's Chi-squared test
## 14 ra        0.107      0.74         1 Pearson's Chi-squared test
## 15 cidi      0.835      0.36         1 Pearson's Chi-squared test
## 16 cgneed    3.18       0.07         1 Pearson's Chi-squared test
##
## [[4]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married  0.474       0.49         1 Pearson's Chi-squared test
##  2 educ     3.22        0.2          2 Pearson's Chi-squared test
##  3 pov      5.56        0.23         4 Pearson's Chi-squared test
##  4 employ   0.495       0.48         1 Pearson's Chi-squared test
##  5 open     1.12        0.29         1 Pearson's Chi-squared test
##  6 race     0.315       0.96         3 Pearson's Chi-squared test
##  7 chdage   0.725       0.87         3 Pearson's Chi-squared test
##  8 cgage    0.299       0.96         3 Pearson's Chi-squared test
##  9 CRA47A   0.466       0.49         1 Pearson's Chi-squared test
## 10 mental   0.000123    0.99         1 Pearson's Chi-squared test
## 11 arrest   3.57        0.06         1 Pearson's Chi-squared test
## 12 PSH17A   1.95        0.16         1 Pearson's Chi-squared test
## 13 maltx    1.25        0.87         4 Pearson's Chi-squared test
## 14 ra       0.108       0.74         1 Pearson's Chi-squared test
## 15 cidi     0.141       0.71         1 Pearson's Chi-squared test
## 16 cgneed   4.11        0.04         1 Pearson's Chi-squared test
##
## [[5]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married     0.160    0.69         1 Pearson's Chi-squared test
##  2 educ        2.66     0.26         2 Pearson's Chi-squared test
##  3 pov         1.57     0.81         4 Pearson's Chi-squared test
##  4 employ      0.623    0.43         1 Pearson's Chi-squared test
##  5 open        5.30     0.02         1 Pearson's Chi-squared test
##  6 race        1.98     0.58         3 Pearson's Chi-squared test
##  7 chdage      2.88     0.41         3 Pearson's Chi-squared test
##  8 cgage       0.848    0.84         3 Pearson's Chi-squared test
##  9 CRA47A      0.203    0.65         1 Pearson's Chi-squared test
## 10 mental      3.86     0.05         1 Pearson's Chi-squared test
## 11 arrest      0.244    0.62         1 Pearson's Chi-squared test
## 12 PSH17A      3.19     0.07         1 Pearson's Chi-squared test
## 13 maltx       2.79     0.59         4 Pearson's Chi-squared test
## 14 ra         19.8      0            1 Pearson's Chi-squared test
## 15 cidi        9.98     0            1 Pearson's Chi-squared test
## 16 cgneed      4.47     0.03         1 Pearson's Chi-squared test
```

We can summarize the results of the balance checks by looking at the significant differences only: In Subclass 1, all participants on "risk assessment" fall into the category of "absence," and on "caregiver report of need," they fall into the category of "no need":

```r
subs %>% filter(q == 1) %>% count(ra)
```

```
## # A tibble: 1 x 3
## # Groups:   q [1]
##       q    ra     n
##   <int> <dbl> <int>
## 1     1     0   545
```

```r
subs %>% filter(q == 1) %>% count(cgneed)
```

```
## # A tibble: 1 x 3
## # Groups:   q [1]
##       q cgneed     n
##   <int>  <dbl> <int>
## 1     1      0   545
```

In Subclass 2, all participants on "risk assessment" fall into the category of "absence," and the chi-square test on "caregiver report of need" is statistically significant ($p < .000$):

```r
subs %>% filter(q == 2) %>% count(ra)
```

```
## # A tibble: 1 x 3
## # Groups:   q [1]
##       q    ra     n
##   <int> <dbl> <int>
## 1     2     0   545
```

```r
subs %>% filter(q == 2) %>% count(cgneed)
```

```
## # A tibble: 2 x 3
## # Groups:   q [1]
##       q cgneed     n
##   <int>  <dbl> <int>
## 1     2      0   543
## 2     2      1     2
```

In Subclass 3, the chi-square test on "AOD treatment receipt" is statistically significant ($p < .01$); in Subclass 4, the chi-square test on "caregiver report of need" is statistically significant ($p < .05$); and in Subclass 5, the chi-square tests on the following covariates are statistically significant: "case status" ($p < .05$), "caregiver mental health problem" ($p < .05$), "risk assessment" ($p < .001$), "CIDI-SF" ($p < .01$), and "caregiver report of need" ($p < .05$).

Trimming

Search for Best Cutoff Alpha for Overlap

As the above balance checks show, the dataset violates the overlap assumption. In other words, there may be parts of the covariate space with limited number of observations for either the treatment or control group. To address the problem of limited overlap, we can follow Crump et al. (2009) and pursue a trimming strategy. This involves searching for a cutoff value and then discarding participants whose estimated propensity scores are below the cutoff value or above one minus the cutoff value. Alternatively, Crump et al., 2009 suggests that a rule of thumb is to use .1 as the cutoff value and to discard all observations with estimated propensity scores outside the range [0.1, 0.9].

```r
# Program to search best cutoff value for overlap (Crump et al., 2009)
trim <- PSweight::PStrim(
  data = as.data.frame(df.ps), # fix for latest package update
  ps.formula = f,
```

```r
  optimal = T
)

# The rule of thumb of .1 approximates the formula for optimal trimming
trim$delta # lower bound (or use .079 from the Stata results)
```

```
## [1] 0.07802586
```

```r
1 - trim$delta # upper bound (or use .921 from the Stata results)
```

```
## [1] 0.9219741
```

Trim Observations

Now that we have found the optimal cutoff point for trimming, we can trim/discard participants whose estimated propensity scores are below .079 and above .921 (i.e., 1 - .079 = .921) from the analysis. This trimming reduces the sample size from 2,723 to 745, or a reduction of 72.6% of the original sample. The sample reduction through trimming is necessary to address the limited overlap problem, but the loss of sample participants is huge. It clearly reduces the study's external validity. This is a limitation of trimming.

```r
summary(df.ps$ps)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.002628 0.014190 0.029990 0.107969 0.093391 0.950519
```

```r
df.trimmed <- df.ps %>%
  filter(ps >= .079 & ps <= .921)
summary(df.trimmed$ps)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.07916 0.13539 0.24566 0.32245 0.47885 0.90917
```

```r
nrow(df.ps)
```

```
## [1] 2723
```

```r
nrow(df.trimmed)
```

```
## [1] 745
```

```r
nrow(df.ps) - nrow(df.trimmed) # trimmed 1,978 obs
```

```
## [1] 1978
```

Create Five New Subclasses and Check Balance

Results of the balance check, shown below, are good. Only three covariates remain statistically significant (i.e., "child age" in Subclass 1, "poverty" in Subclass 3, and "CIDI-SF" in Subclass 5), and all other covariates are not statistically significant. Note that the significance of "child age" may be interpreted as chance significance, because it was never significant in all previous schemes.

```r
# Cut 5 subclasses (Type 2 and 5 in quantile() are identical to the Stata results)
subs_t <- df.trimmed %>%
  arrange(ps) %>%
  group_by(q = cut(
    ps,
    include.lowest = F,
    right = F,
    dig.lab = 7,
```

```r
    breaks = c(0, round(quantile(df.trimmed$ps,
      c(.2, .4, .6, .8),
      type = 2
    ), 7), 2),
    labels = F
  ))

# Summary statistics of ps per subclass
subs_t %>%
  group_by(q) %>%
  summarise(
    obs = n(),
    mean = mean(ps),
    sd = sd(ps),
    min = min(ps),
    max = max(ps), .groups = "drop"
  ) %>%
  left_join(
    subs %>%
      count(q, aodserv) %>%
      pivot_wider(names_from = "aodserv", values_from = "n", names_prefix = "aodserv"),
    by = "q"
  )
```

```
## # A tibble: 5 x 8
##       q   obs   mean     sd    min   max aodserv0 aodserv1
##   <int> <int>  <dbl>  <dbl>  <dbl> <dbl>    <int>    <int>
## 1     1   149 0.0959 0.0102 0.0792 0.115      544        1
## 2     2   149 0.151  0.0212 0.115  0.190      539        6
## 3     3   149 0.249  0.0386 0.190  0.317      529       14
## 4     4   149 0.422  0.0675 0.318  0.546      496       50
## 5     5   149 0.694  0.102  0.549  0.909      321      223
```

```r
# Check balance
subs_t %>%
  group_map(~ map_dfr(covs2, check_blc, df = .x))
```

```
## [[1]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married   3.55       0.06         1 Pearson's Chi-squared test
##  2 educ      3.05       0.22         2 Pearson's Chi-squared test
##  3 pov       2.80       0.59         4 Pearson's Chi-squared test
##  4 employ    0.697      0.4          1 Pearson's Chi-squared test
##  5 open      1.28       0.26         1 Pearson's Chi-squared test
##  6 race      1.50       0.68         3 Pearson's Chi-squared test
##  7 chdage   11.6        0.01         3 Pearson's Chi-squared test
##  8 cgage     0.346      0.95         3 Pearson's Chi-squared test
##  9 CRA47A    0.158      0.69         1 Pearson's Chi-squared test
## 10 mental    0.119      0.73         1 Pearson's Chi-squared test
## 11 arrest    0.0235     0.88         1 Pearson's Chi-squared test
## 12 PSH17A    1.14       0.29         1 Pearson's Chi-squared test
## 13 maltx     1.52       0.82         4 Pearson's Chi-squared test
## 14 ra        0.146      0.7          1 Pearson's Chi-squared test
```

```
## 15 cidi      0.00627    0.94         1 Pearson's Chi-squared test
## 16 cgneed    0.810      0.37         1 Pearson's Chi-squared test
##
## [[2]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married   1.15       0.28         1 Pearson's Chi-squared test
##  2 educ      3.50       0.17         2 Pearson's Chi-squared test
##  3 pov       1.45       0.84         4 Pearson's Chi-squared test
##  4 employ    0.882      0.35         1 Pearson's Chi-squared test
##  5 open      2.79       0.1          1 Pearson's Chi-squared test
##  6 race      0.320      0.96         3 Pearson's Chi-squared test
##  7 chdage    1.45       0.69         3 Pearson's Chi-squared test
##  8 cgage     5.59       0.13         3 Pearson's Chi-squared test
##  9 CRA47A    0.00630    0.94         1 Pearson's Chi-squared test
## 10 mental    0.276      0.6          1 Pearson's Chi-squared test
## 11 arrest    3.07       0.08         1 Pearson's Chi-squared test
## 12 PSH17A    0.903      0.34         1 Pearson's Chi-squared test
## 13 maltx     3.60       0.46         4 Pearson's Chi-squared test
## 14 ra        0.0639     0.8          1 Pearson's Chi-squared test
## 15 cidi      0.563      0.45         1 Pearson's Chi-squared test
## 16 cgneed    0.00555    0.94         1 Pearson's Chi-squared test
##
## [[3]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married   0.0707     0.79         1 Pearson's Chi-squared test
##  2 educ      5.38       0.07         2 Pearson's Chi-squared test
##  3 pov      14.5        0.01         4 Pearson's Chi-squared test
##  4 employ    1.15       0.28         1 Pearson's Chi-squared test
##  5 open      0.0195     0.89         1 Pearson's Chi-squared test
##  6 race      1.05       0.79         3 Pearson's Chi-squared test
##  7 chdage    1.11       0.77         3 Pearson's Chi-squared test
##  8 cgage     0.0411     1            3 Pearson's Chi-squared test
##  9 CRA47A    0.00458    0.95         1 Pearson's Chi-squared test
## 10 mental    1.37       0.24         1 Pearson's Chi-squared test
## 11 arrest    3.34       0.07         1 Pearson's Chi-squared test
## 12 PSH17A    0.0215     0.88         1 Pearson's Chi-squared test
## 13 maltx     2.30       0.68         4 Pearson's Chi-squared test
## 14 ra        1.30       0.25         1 Pearson's Chi-squared test
## 15 cidi      1.02       0.31         1 Pearson's Chi-squared test
## 16 cgneed    0.103      0.75         1 Pearson's Chi-squared test
##
## [[4]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married    0.381     0.54         1 Pearson's Chi-squared test
##  2 educ       4.39      0.11         2 Pearson's Chi-squared test
##  3 pov        6.41      0.17         4 Pearson's Chi-squared test
##  4 employ     0.475     0.49         1 Pearson's Chi-squared test
##  5 open       0.136     0.71         1 Pearson's Chi-squared test
```

```
##  6 race       0.471     0.93         3 Pearson's Chi-squared test
##  7 chdage     0.640     0.89         3 Pearson's Chi-squared test
##  8 cgage      0.196     0.98         3 Pearson's Chi-squared test
##  9 CRA47A     1.84      0.18         1 Pearson's Chi-squared test
## 10 mental     2.93      0.09         1 Pearson's Chi-squared test
## 11 arrest     0.147     0.7          1 Pearson's Chi-squared test
## 12 PSH17A     0.134     0.71         1 Pearson's Chi-squared test
## 13 maltx      1.60      0.81         4 Pearson's Chi-squared test
## 14 ra         1.00      0.32         1 Pearson's Chi-squared test
## 15 cidi       0.0107    0.92         1 Pearson's Chi-squared test
## 16 cgneed     3.79      0.05         1 Pearson's Chi-squared test
##
## [[5]]
## # A tibble: 16 x 5
##    cov     statistic p.value parameter method
##    <chr>       <dbl>   <dbl>     <int> <chr>
##  1 married     0.766    0.38         1 Pearson's Chi-squared test
##  2 educ        3.03     0.22         2 Pearson's Chi-squared test
##  3 pov         1.35     0.85         4 Pearson's Chi-squared test
##  4 employ      0.335    0.56         1 Pearson's Chi-squared test
##  5 open        0.105    0.75         1 Pearson's Chi-squared test
##  6 race        3.13     0.37         3 Pearson's Chi-squared test
##  7 chdage      1.70     0.64         3 Pearson's Chi-squared test
##  8 cgage       1.02     0.8          3 Pearson's Chi-squared test
##  9 CRA47A      0.698    0.4          1 Pearson's Chi-squared test
## 10 mental      0.108    0.74         1 Pearson's Chi-squared test
## 11 arrest      1.15     0.28         1 Pearson's Chi-squared test
## 12 PSH17A      0.332    0.56         1 Pearson's Chi-squared test
## 13 maltx       5.93     0.2          4 Pearson's Chi-squared test
## 14 ra          0.116    0.73         1 Pearson's Chi-squared test
## 15 cidi        4.26     0.04         1 Pearson's Chi-squared test
## 16 cgneed      2.54     0.11         1 Pearson's Chi-squared test
```

```r
# Filter significant results
subs_t %>%
  group_map(~ map_dfr(covs2, check_blc, df = .x) %>% filter(p.value < .05))
```

```
## [[1]]
## # A tibble: 1 x 5
##   cov    statistic p.value parameter method
##   <chr>      <dbl>   <dbl>     <int> <chr>
## 1 chdage      11.6    0.01         3 Pearson's Chi-squared test
##
## [[2]]
## # A tibble: 0 x 5
## # i 5 variables: cov <chr>, statistic <dbl>, p.value <dbl>, parameter <int>,
## #   method <chr>
##
## [[3]]
## # A tibble: 1 x 5
##   cov   statistic p.value parameter method
##   <chr>     <dbl>   <dbl>     <int> <chr>
## 1 pov        14.5    0.01         4 Pearson's Chi-squared test
##
## [[4]]
```

```
## # A tibble: 0 x 5
## # i 5 variables: cov <chr>, statistic <dbl>, p.value <dbl>, parameter <int>,
## #   method <chr>
##
## [[5]]
## # A tibble: 1 x 5
##   cov   statistic p.value parameter method
##   <chr>     <dbl>   <dbl>     <int> <chr>
## 1 cidi       4.26    0.04         1 Pearson's Chi-squared test
```

Outcome Analysis by Subclass

```
(f2 <- as.formula(paste0(
  "Surv(durm, rrpt) ~ aodserv + married + high + bahigh + ",
  "poverty2 + poverty3 + poverty4 + poverty5 + employ + open + black + ",
  "hispanic + natam + chdage1 + chdage2 + chdage3 + CRA47A"
)))
```

```
## Surv(durm, rrpt) ~ aodserv + married + high + bahigh + poverty2 +
##     poverty3 + poverty4 + poverty5 + employ + open + black +
##     hispanic + natam + chdage1 + chdage2 + chdage3 + CRA47A
```

```
subs_t %>%
  group_map(~ broom::tidy(survival::coxph(f2, .x)))
```

```
## [[1]]
## # A tibble: 17 x 5
##    term     estimate std.error statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv    1.28       0.523     2.45   0.0143
##  2 married    0.0453     0.435     0.104  0.917
##  3 high      -0.223      0.450    -0.496  0.620
##  4 bahigh    -0.580      0.572    -1.01   0.310
##  5 poverty2  -0.245      0.488    -0.503  0.615
##  6 poverty3   0.748      0.535     1.40   0.162
##  7 poverty4   0.306      0.669     0.456  0.648
##  8 poverty5  -1.15       1.11     -1.03   0.301
##  9 employ     0.0966     0.414     0.233  0.816
## 10 open       0.339      0.520     0.652  0.515
## 11 black      0.428      0.465     0.921  0.357
## 12 hispanic   0.441      0.606     0.728  0.467
## 13 natam      0.372      0.625     0.595  0.552
## 14 chdage1   -1.40       0.551    -2.55   0.0109
## 15 chdage2   -1.19       0.763    -1.56   0.119
## 16 chdage3   -0.924      0.555    -1.67   0.0955
## 17 CRA47A     0.665      0.427     1.56   0.119
##
## [[2]]
## # A tibble: 17 x 5
##    term     estimate std.error statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv   0.0990      0.443    0.224   0.823
##  2 married  -0.0473      0.403   -0.117   0.906
##  3 high      0.392       0.385    1.02    0.309
##  4 bahigh    0.489       0.500    0.979   0.328
```

```
##  5 poverty2  0.369       0.417    0.884   0.377
##  6 poverty3 -2.09        1.08    -1.92    0.0544
##  7 poverty4  0.0632      0.580    0.109   0.913
##  8 poverty5 -1.25        1.10    -1.14    0.256
##  9 employ    0.251       0.358    0.701   0.483
## 10 open     -0.345       0.379   -0.910   0.363
## 11 black    -0.123       0.496   -0.247   0.805
## 12 hispanic  0.307       0.456    0.672   0.501
## 13 natam     0.390       0.586    0.666   0.505
## 14 chdage1  -0.162       0.472   -0.343   0.732
## 15 chdage2  -0.544       0.657   -0.827   0.408
## 16 chdage3   0.342       0.555    0.616   0.538
## 17 CRA47A    0.00356     0.335    0.0106  0.992
##
## [[3]]
## # A tibble: 17 x 5
##    term     estimate std.error statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv   -0.168      0.449   -0.375   0.708
##  2 married   -0.800      0.561   -1.43    0.154
##  3 high      -0.0488     0.452   -0.108   0.914
##  4 bahigh    -0.188      0.570   -0.329   0.742
##  5 poverty2   0.0219     0.486    0.0451  0.964
##  6 poverty3  -0.309      0.692   -0.447   0.655
##  7 poverty4  -0.281      0.844   -0.332   0.740
##  8 poverty5  -0.586      0.702   -0.836   0.403
##  9 employ    -0.341      0.443   -0.772   0.440
## 10 open       0.259      0.440    0.589   0.556
## 11 black     -0.500      0.460   -1.09    0.277
## 12 hispanic  -1.24       0.679   -1.82    0.0687
## 13 natam     -0.118      0.735   -0.160   0.873
## 14 chdage1   -0.0163     0.504   -0.0324  0.974
## 15 chdage2    0.554      0.716    0.774   0.439
## 16 chdage3   -0.486      0.700   -0.694   0.488
## 17 CRA47A     0.664      0.382    1.74    0.0820
##
## [[4]]
## # A tibble: 17 x 5
##    term     estimate std.error statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv    0.245      0.377    0.649    0.516
##  2 married    0.694      0.428    1.62     0.105
##  3 high       0.512      0.426    1.20     0.230
##  4 bahigh    -0.323      0.711   -0.454    0.650
##  5 poverty2   0.315      0.478    0.658    0.510
##  6 poverty3  -0.132      0.672   -0.196    0.844
##  7 poverty4   0.350      0.928    0.377    0.706
##  8 poverty5  -0.988      0.855   -1.16     0.248
##  9 employ    -0.301      0.393   -0.767    0.443
## 10 open       0.361      0.484    0.746    0.456
## 11 black     -0.0424     0.457   -0.0927   0.926
## 12 hispanic   0.0690     0.664    0.104    0.917
## 13 natam     -1.66       1.07    -1.56     0.119
## 14 chdage1   -0.644      0.520   -1.24     0.216
```

```
## 15 chdage2    0.318      0.602    0.528    0.598
## 16 chdage3   -0.406      0.568   -0.715    0.474
## 17 CRA47A     0.569      0.402    1.42     0.157
##
## [[5]]
## # A tibble: 17 x 5
##    term     estimate std.error statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv    0.419      0.403     1.04   0.298
##  2 married    0.116      0.449     0.258  0.796
##  3 high      -0.696      0.425    -1.64   0.101
##  4 bahigh    -1.13       0.649    -1.74   0.0824
##  5 poverty2   0.910      0.402     2.27   0.0235
##  6 poverty3  -1.18       1.09     -1.09   0.276
##  7 poverty4  -0.609      0.818    -0.744  0.457
##  8 poverty5   0.0969     0.642     0.151  0.880
##  9 employ     0.127      0.405     0.313  0.755
## 10 open       0.492      0.638     0.772  0.440
## 11 black     -1.07       0.466    -2.29   0.0220
## 12 hispanic  -0.526      0.514    -1.02   0.306
## 13 natam     -0.107      0.544    -0.197  0.844
## 14 chdage1   -0.407      0.665    -0.613  0.540
## 15 chdage2    0.872      0.694     1.26   0.209
## 16 chdage3    0.491      0.729     0.674  0.500
## 17 CRA47A     0.0749     0.359     0.209  0.835
```

Aggregate Results

Extracting the coefficients of receipt of substance abuse treatment and the corresponding standard errors from all five subclasses, the study then computes the overall effect by aggregating the five estimated coefficients and computes the associated standard error by aggregating the five estimated standard errors. The purpose of this analysis is to discern whether the treatment effect across all five subclasses is statistically significant.

The weighted average of the within-stratum estimates, given K strata, the sample size per strata n_k, the total sample size N, and the stratum ATE $\hat{\tau}_k$, is:

$$\hat{\tau} = \sum_{k=1}^{K} \frac{n_k}{N} \hat{\tau}_k$$

And the variance of the sample ATE is estimated by:

$$Var(\hat{\tau}) = \sum_{k=1}^{K} \left(\frac{n_k}{N}\right)^2 \text{Var}\left[\bar{Y}_{0k} - \bar{Y}_{1k}\right]$$

We can then perform a significance test of nondirectional or directional hypothesis via the square root of the variance or the standard error

$$SE(\hat{\tau}) = \sqrt{\text{Var}(\hat{\tau})}$$

```r
# Aggregate the five treatment effects
(subs_coefs <- subs_t %>%
  group_map(~ broom::tidy(survival::coxph(f2, .x)) %>%
```

```r
              filter(term == "aodserv") %>% pull(estimate)) %>%
  unlist())
```

```
## [1]  1.28101311  0.09899715 -0.16800177  0.24482769  0.41883025
```

```r
(subs_n <- subs_t %>% count(q) %>% pull(n))
```

```
## [1] 149 149 149 149 149
```

```r
(subs_ate <- sum(subs_coefs * (subs_n / sum(subs_n))))
```

```
## [1] 0.3751333
```

```r
# Calculate the hazard ratio
exp(subs_ate)
```

```
## [1] 1.455185
```

```r
# Aggregate the variance
(subs_vars <- subs_t %>%
  group_map(~ broom::tidy(survival::coxph(f2, .x)) %>%
              filter(term == "aodserv") %>% pull(std.error)) %>%
  unlist())
```

```
## [1] 0.5228186 0.4428462 0.4485225 0.3773371 0.4027468
```

```r
(subs_se <- sqrt(sum(subs_vars^2 * (subs_n / sum(subs_n))^2)))
```

```
## [1] 0.1975057
```

```r
# Calculate z-score and two-sided p-value
(z <- subs_ate / subs_se)
```

```
## [1] 1.899354
```

```r
(p.value <- (1 - pnorm(z)) * 2)
```

```
## [1] 0.05751794
```

Results with Subclassification: The study finds that by controlling for selection bias, children whose caregivers used substance abuse treatment are more likely to receive a maltreatment re-report, and the hazard of having a re-report for these children is 45.52% higher than that of children whose caregivers did not use substance abuse treatment, although this difference is not statistically significant at the .05 level ($p = .0575$).

Results without Subclassification: The Cox regression without controlling for selection bias shows that the hazard ratio of treatment is 1.55 (i.e., $\exp(0.436) = 1.55$), close to that estimated by the subclassified Cox regression, but the finding is statistically significant at the .001 level. The statistical significance of the treatment effect from the "overall" model appears influenced by uncontrolled selectivity:

```r
# Cox regression without controlling for selection bias
broom::tidy(survival::coxph(f2, df)) %>%
  filter(term == "aodserv") %>%
  mutate(hazard.ratio = exp(estimate))
```

```
## # A tibble: 1 x 6
##   term    estimate std.error statistic  p.value hazard.ratio
##   <chr>      <dbl>     <dbl>     <dbl>    <dbl>        <dbl>
## 1 aodserv    0.436     0.123      3.54 0.000404         1.55
```

Subclassification with MatchIt

The `MatchIt` package provides a convenient interface for performing subclassification without relying on any other packages.[5] Instead of pooling the stratum-specific treatment effects, it produces stratum weights that can be used to estimate a single marginal effect. This approach is known as marginal mean weighting through stratification (MMWS) (Hong, 2010).[6] The robust standard error (`robust = T`) must be used because of the MMWS weights. Note also that the `discard` argument contains options for discarding units outside a region of common support.

```r
# Without trimming
(mm <- MatchIt::matchit(f, df, method = "subclass", subclass = 5))
```

```
## A matchit object
##  - method: Subclassification (5 subclasses)
##  - distance: Propensity score
##              - estimated with logistic regression
##  - number of obs.: 2723 (original), 2723 (matched)
##  - target estimand: ATT
##  - covariates: married, high, bahigh, poverty2, poverty3, poverty4, poverty5, employ, open, black, hi
```

```r
table(mm$subclass)
```

```
##
##    1    2    3    4    5
## 2039  345  155   98   86
```

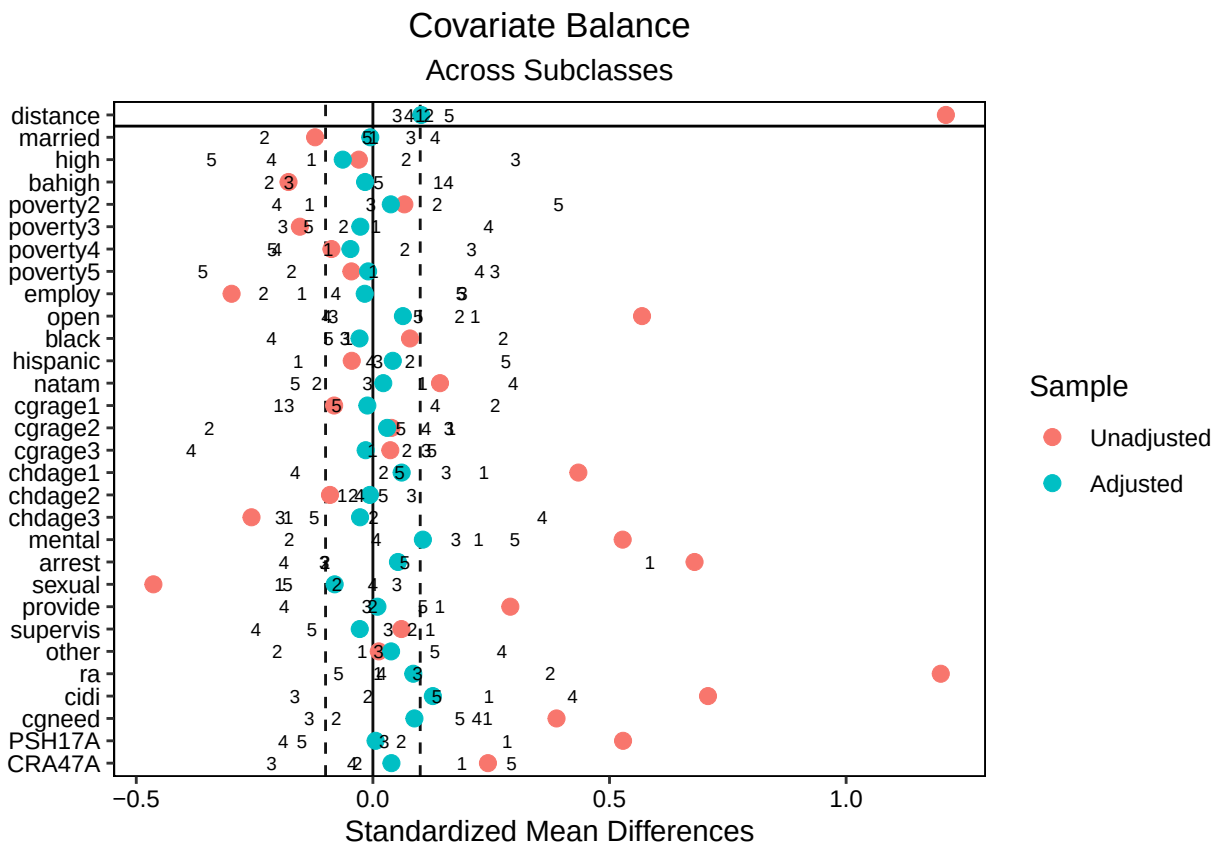
```r
mm$q.cut # vector of propensity score cutpoints to define subclasses
```

```
##         0%        20%        40%        60%        80%       100%
## 0.01097901 0.09224427 0.28514176 0.49620042 0.65903538 0.94833436
```

```r
cobalt::love.plot(mm, binary = "std", disp.subclass = T, threshold = c(m = .1))
```

[5] https://kosukeimai.github.io/MatchIt/reference/method_subclass.html

[6] Hong, G. (2010). Marginal Mean Weighting Through Stratification: Adjustment for Selection Bias in Multilevel Data. *Journal of Educational and Behavioral Statistics*, *35*(5), 499–531. https://doi.org/10.3102/1076998609359785




```
cobalt::bal.tab(mm, un = T, binary = "std", disp.subclass = T, threshold = c(m = .1))
```

```
## Balance by subclass
##  - - - Subclass 1 - - -
##               Type Diff.Adj        M.Threshold
## distance Distance   0.1004
## married    Binary   0.0019     Balanced, <0.1
## high       Binary  -0.1296 Not Balanced, >0.1
## bahigh     Binary   0.1385 Not Balanced, >0.1
## poverty2   Binary  -0.1337 Not Balanced, >0.1
## poverty3   Binary   0.0069     Balanced, <0.1
## poverty4   Binary  -0.0942     Balanced, <0.1
## poverty5   Binary   0.0016     Balanced, <0.1
## employ     Binary  -0.1496 Not Balanced, >0.1
## open       Binary   0.2167 Not Balanced, >0.1
## black      Binary  -0.0522     Balanced, <0.1
## hispanic   Binary  -0.1573 Not Balanced, >0.1
## natam      Binary   0.1054 Not Balanced, >0.1
## cgrage1    Binary  -0.1987 Not Balanced, >0.1
## cgrage2    Binary   0.1653 Not Balanced, >0.1
## cgrage3    Binary  -0.0007     Balanced, <0.1
## chdage1    Binary   0.2348 Not Balanced, >0.1
## chdage2    Binary  -0.0648     Balanced, <0.1
## chdage3    Binary  -0.1786 Not Balanced, >0.1
## mental     Binary   0.2233 Not Balanced, >0.1
## arrest     Binary   0.5853 Not Balanced, >0.1
## sexual     Binary  -0.1967 Not Balanced, >0.1
```

```
## provide    Binary   0.1416 Not Balanced, >0.1
## supervis   Binary   0.1216 Not Balanced, >0.1
## other      Binary  -0.0225     Balanced, <0.1
## ra         Binary   0.0114     Balanced, <0.1
## cidi       Binary   0.2453 Not Balanced, >0.1
## cgneed     Binary   0.2427 Not Balanced, >0.1
## PSH17A     Binary   0.2839 Not Balanced, >0.1
## CRA47A     Binary   0.1883 Not Balanced, >0.1
##
##  - - - Subclass 2 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.1188
## married    Binary  -0.2288 Not Balanced, >0.1
## high       Binary   0.0708     Balanced, <0.1
## bahigh     Binary  -0.2186 Not Balanced, >0.1
## poverty2   Binary   0.1363 Not Balanced, >0.1
## poverty3   Binary  -0.0619     Balanced, <0.1
## poverty4   Binary   0.0678     Balanced, <0.1
## poverty5   Binary  -0.1715 Not Balanced, >0.1
## employ     Binary  -0.2305 Not Balanced, >0.1
## open       Binary   0.1832 Not Balanced, >0.1
## black      Binary   0.2759 Not Balanced, >0.1
## hispanic   Binary   0.0786     Balanced, <0.1
## natam      Binary  -0.1183 Not Balanced, >0.1
## cgrage1    Binary   0.2587 Not Balanced, >0.1
## cgrage2    Binary  -0.3458 Not Balanced, >0.1
## cgrage3    Binary   0.0723     Balanced, <0.1
## chdage1    Binary   0.0229     Balanced, <0.1
## chdage2    Binary  -0.0415     Balanced, <0.1
## chdage3    Binary   0.0015     Balanced, <0.1
## mental     Binary  -0.1763 Not Balanced, >0.1
## arrest     Binary  -0.1005 Not Balanced, >0.1
## sexual     Binary  -0.0758     Balanced, <0.1
## provide    Binary  -0.0004     Balanced, <0.1
## supervis   Binary   0.0833     Balanced, <0.1
## other      Binary  -0.2021 Not Balanced, >0.1
## ra         Binary   0.3748 Not Balanced, >0.1
## cidi       Binary  -0.0102     Balanced, <0.1
## cgneed     Binary  -0.0775     Balanced, <0.1
## PSH17A     Binary   0.0598     Balanced, <0.1
## CRA47A     Binary  -0.0331     Balanced, <0.1
##
##  - - - Subclass 3 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0517
## married    Binary   0.0807     Balanced, <0.1
## high       Binary   0.3018 Not Balanced, >0.1
## bahigh     Binary  -0.1770 Not Balanced, >0.1
## poverty2   Binary  -0.0041     Balanced, <0.1
## poverty3   Binary  -0.1898 Not Balanced, >0.1
## poverty4   Binary   0.2086 Not Balanced, >0.1
## poverty5   Binary   0.2580 Not Balanced, >0.1
## employ     Binary   0.1896 Not Balanced, >0.1
## open       Binary  -0.0839     Balanced, <0.1
```

```
## black      Binary  -0.0586     Balanced, <0.1
## hispanic   Binary   0.0110     Balanced, <0.1
## natam      Binary  -0.0113     Balanced, <0.1
## cgrage1    Binary  -0.1763 Not Balanced, >0.1
## cgrage2    Binary   0.1605 Not Balanced, >0.1
## cgrage3    Binary   0.1128 Not Balanced, >0.1
## chdage1    Binary   0.1550 Not Balanced, >0.1
## chdage2    Binary   0.0819     Balanced, <0.1
## chdage3    Binary  -0.1957 Not Balanced, >0.1
## mental     Binary   0.1753 Not Balanced, >0.1
## arrest     Binary  -0.1023 Not Balanced, >0.1
## sexual     Binary   0.0510     Balanced, <0.1
## provide    Binary  -0.0124     Balanced, <0.1
## supervis   Binary   0.0325     Balanced, <0.1
## other      Binary   0.0121     Balanced, <0.1
## ra         Binary   0.0944     Balanced, <0.1
## cidi       Binary  -0.1646 Not Balanced, >0.1
## cgneed     Binary  -0.1342 Not Balanced, >0.1
## PSH17A     Binary   0.0243     Balanced, <0.1
## CRA47A     Binary  -0.2136 Not Balanced, >0.1
##
##  - - - Subclass 4 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0773
## married    Binary   0.1316 Not Balanced, >0.1
## high       Binary  -0.2141 Not Balanced, >0.1
## bahigh     Binary   0.1601 Not Balanced, >0.1
## poverty2   Binary  -0.2027 Not Balanced, >0.1
## poverty3   Binary   0.2446 Not Balanced, >0.1
## poverty4   Binary  -0.2026 Not Balanced, >0.1
## poverty5   Binary   0.2254 Not Balanced, >0.1
## employ     Binary  -0.0780     Balanced, <0.1
## open       Binary  -0.0974     Balanced, <0.1
## black      Binary  -0.2136 Not Balanced, >0.1
## hispanic   Binary  -0.0038     Balanced, <0.1
## natam      Binary   0.2969 Not Balanced, >0.1
## cgrage1    Binary   0.1318 Not Balanced, >0.1
## cgrage2    Binary   0.1132 Not Balanced, >0.1
## cgrage3    Binary  -0.3836 Not Balanced, >0.1
## chdage1    Binary  -0.1638 Not Balanced, >0.1
## chdage2    Binary  -0.0279     Balanced, <0.1
## chdage3    Binary   0.3581 Not Balanced, >0.1
## mental     Binary   0.0070     Balanced, <0.1
## arrest     Binary  -0.1881 Not Balanced, >0.1
## sexual     Binary   0.0000     Balanced, <0.1
## provide    Binary  -0.1871 Not Balanced, >0.1
## supervis   Binary  -0.2469 Not Balanced, >0.1
## other      Binary   0.2729 Not Balanced, >0.1
## ra         Binary   0.0185     Balanced, <0.1
## cidi       Binary   0.4220 Not Balanced, >0.1
## cgneed     Binary   0.2198 Not Balanced, >0.1
## PSH17A     Binary  -0.1891 Not Balanced, >0.1
## CRA47A     Binary  -0.0432     Balanced, <0.1
##
```

```
##  - - - Subclass 5 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.1617
## married    Binary  -0.0115     Balanced, <0.1
## high       Binary  -0.3409 Not Balanced, >0.1
## bahigh     Binary   0.0120     Balanced, <0.1
## poverty2   Binary   0.3926 Not Balanced, >0.1
## poverty3   Binary  -0.1355 Not Balanced, >0.1
## poverty4   Binary  -0.2126 Not Balanced, >0.1
## poverty5   Binary  -0.3593 Not Balanced, >0.1
## employ     Binary   0.1860 Not Balanced, >0.1
## open       Binary   0.0958     Balanced, <0.1
## black      Binary  -0.0931     Balanced, <0.1
## hispanic   Binary   0.2813 Not Balanced, >0.1
## natam      Binary  -0.1636 Not Balanced, >0.1
## cgrage1    Binary  -0.0767     Balanced, <0.1
## cgrage2    Binary   0.0593     Balanced, <0.1
## cgrage3    Binary   0.1251 Not Balanced, >0.1
## chdage1    Binary   0.0566     Balanced, <0.1
## chdage2    Binary   0.0220     Balanced, <0.1
## chdage3    Binary  -0.1238 Not Balanced, >0.1
## mental     Binary   0.3001 Not Balanced, >0.1
## arrest     Binary   0.0676     Balanced, <0.1
## sexual     Binary  -0.1802 Not Balanced, >0.1
## provide    Binary   0.1060 Not Balanced, >0.1
## supervis   Binary  -0.1284 Not Balanced, >0.1
## other      Binary   0.1314 Not Balanced, >0.1
## ra         Binary  -0.0723     Balanced, <0.1
## cidi       Binary   0.1358 Not Balanced, >0.1
## cgneed     Binary   0.1843 Not Balanced, >0.1
## PSH17A     Binary  -0.1499 Not Balanced, >0.1
## CRA47A     Binary   0.2930 Not Balanced, >0.1
```

```r
broom::tidy(survival::coxph(f2, df, mm$weights, robust = T))
```

```
## # A tibble: 17 x 6
##    term     estimate std.error robust.se statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv    0.434     0.119      0.172    2.51    0.0119
##  2 married    0.102     0.102      0.238    0.429   0.668
##  3 high      -0.102     0.0971     0.221   -0.461   0.645
##  4 bahigh    -0.484     0.150      0.262   -1.85    0.0643
##  5 poverty2   0.242     0.105      0.243    0.992   0.321
##  6 poverty3  -0.623     0.167      0.306   -2.04    0.0415
##  7 poverty4  -0.118     0.168      0.386   -0.306   0.760
##  8 poverty5  -0.398     0.176      0.451   -0.882   0.378
##  9 employ     0.0465    0.0930     0.211    0.221   0.825
## 10 open      -0.0987    0.0989     0.235   -0.420   0.674
## 11 black     -0.493     0.112      0.266   -1.86    0.0634
## 12 hispanic   0.123     0.126      0.264    0.466   0.641
## 13 natam      0.0575    0.156      0.475    0.121   0.904
## 14 chdage1   -0.352     0.118      0.247   -1.42    0.155
## 15 chdage2    0.0165    0.141      0.281    0.0586  0.953
## 16 chdage3   -0.461     0.144      0.236   -1.96    0.0501
## 17 CRA47A     0.411     0.0880     0.201    2.04    0.0410
```

```r
exp(0.434) # significant, but note the violations of the overlap assumption
```

```
## [1] 1.543419
```

```r
# With trimmed dataset & outcome analysis with MMWS
(mm2 <- MatchIt::matchit(f, df.trimmed, method = "subclass", subclass = 5)) # ATT
```

```
## A matchit object
##  - method: Subclassification (5 subclasses)
##  - distance: Propensity score
##              - estimated with logistic regression
##  - number of obs.: 745 (original), 745 (matched)
##  - target estimand: ATT
##  - covariates: married, high, bahigh, poverty2, poverty3, poverty4, poverty5, employ, open, black, hi
```

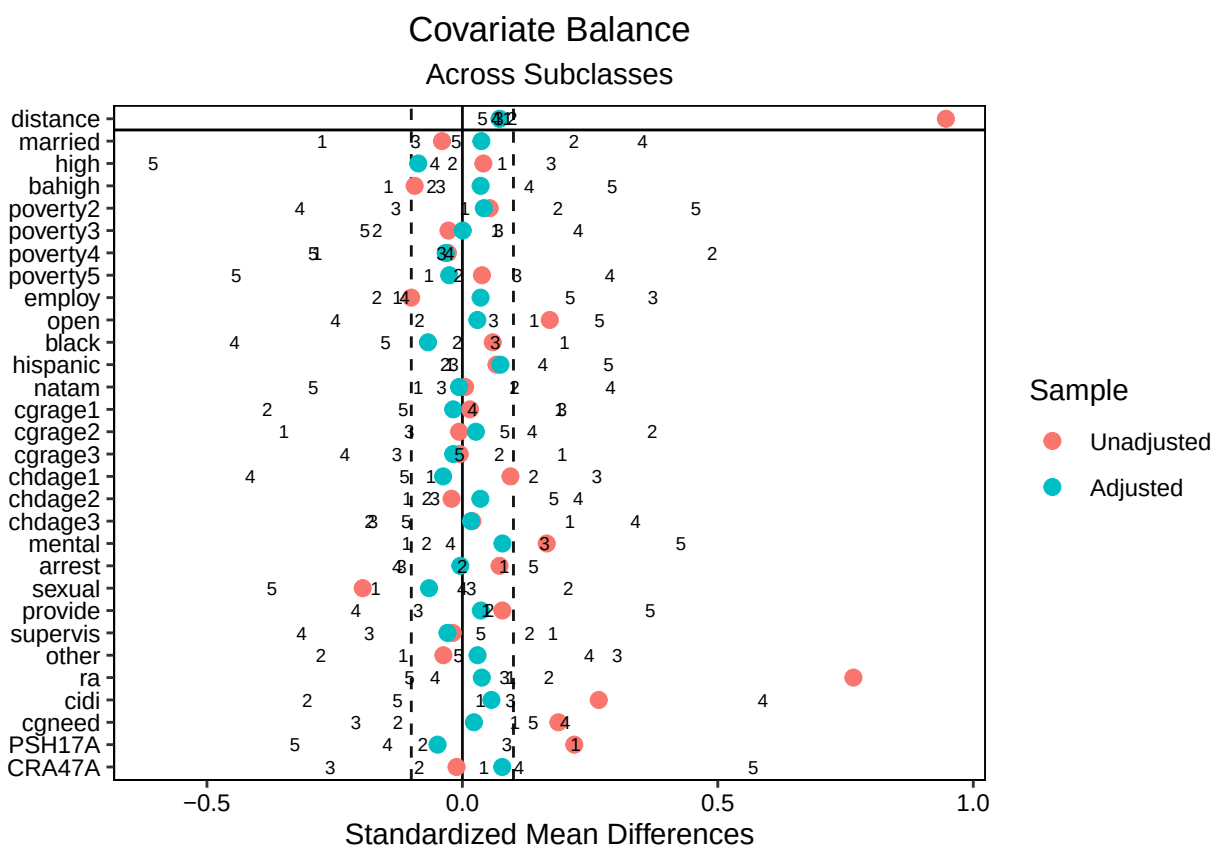
```r
table(mm2$subclass)
```

```
##
##   1   2   3   4   5
## 339 156 104  81  65
```

```r
mm2$q.cut # vector of propensity score cutpoints to define subclasses
```

```
##          0%         20%         40%         60%         80%        100%
## 0.08287899 0.22175950 0.38399142 0.54616331 0.66903253 0.89800621
```

```r
cobalt::love.plot(mm2, binary = "std", disp.subclass = T, threshold = c(m = .1))
```




```r
cobalt::bal.tab(mm2, un = T, binary = "std", disp.subclass = T, threshold = c(m = .1))
```

```
## Balance by subclass
##  - - - Subclass 1 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0887
## married    Binary  -0.2741 Not Balanced, >0.1
## high       Binary   0.0785     Balanced, <0.1
## bahigh     Binary  -0.1446 Not Balanced, >0.1
## poverty2   Binary   0.0050     Balanced, <0.1
## poverty3   Binary   0.0655     Balanced, <0.1
## poverty4   Binary  -0.2841 Not Balanced, >0.1
## poverty5   Binary  -0.0653     Balanced, <0.1
## employ     Binary  -0.1269 Not Balanced, >0.1
## open       Binary   0.1409 Not Balanced, >0.1
## black      Binary   0.2006 Not Balanced, >0.1
## hispanic   Binary  -0.0239     Balanced, <0.1
## natam      Binary  -0.0861     Balanced, <0.1
## cgrage1    Binary   0.1905 Not Balanced, >0.1
## cgrage2    Binary  -0.3493 Not Balanced, >0.1
## cgrage3    Binary   0.1958 Not Balanced, >0.1
## chdage1    Binary  -0.0616     Balanced, <0.1
## chdage2    Binary  -0.1074 Not Balanced, >0.1
## chdage3    Binary   0.2109 Not Balanced, >0.1
## mental     Binary  -0.1083 Not Balanced, >0.1
## arrest     Binary   0.0819     Balanced, <0.1
## sexual     Binary  -0.1696 Not Balanced, >0.1
## provide    Binary   0.0473     Balanced, <0.1
## supervis   Binary   0.1775 Not Balanced, >0.1
## other      Binary  -0.1155 Not Balanced, >0.1
## ra         Binary   0.0953     Balanced, <0.1
## cidi       Binary   0.0361     Balanced, <0.1
## cgneed     Binary   0.1029 Not Balanced, >0.1
## PSH17A     Binary   0.2217 Not Balanced, >0.1
## CRA47A     Binary   0.0426     Balanced, <0.1
##
##  - - - Subclass 2 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0976
## married    Binary   0.2182 Not Balanced, >0.1
## high       Binary  -0.0187     Balanced, <0.1
## bahigh     Binary  -0.0600     Balanced, <0.1
## poverty2   Binary   0.1872 Not Balanced, >0.1
## poverty3   Binary  -0.1662 Not Balanced, >0.1
## poverty4   Binary   0.4894 Not Balanced, >0.1
## poverty5   Binary  -0.0069     Balanced, <0.1
## employ     Binary  -0.1673 Not Balanced, >0.1
## open       Binary  -0.0833     Balanced, <0.1
## black      Binary  -0.0101     Balanced, <0.1
## hispanic   Binary  -0.0333     Balanced, <0.1
## natam      Binary   0.1026 Not Balanced, >0.1
## cgrage1    Binary  -0.3815 Not Balanced, >0.1
## cgrage2    Binary   0.3720 Not Balanced, >0.1
## cgrage3    Binary   0.0725     Balanced, <0.1
## chdage1    Binary   0.1397 Not Balanced, >0.1
## chdage2    Binary  -0.0693     Balanced, <0.1
```

```
## chdage3    Binary  -0.1816 Not Balanced, >0.1
## mental     Binary  -0.0697     Balanced, <0.1
## arrest     Binary   0.0000     Balanced, <0.1
## sexual     Binary   0.2072 Not Balanced, >0.1
## provide    Binary   0.0528     Balanced, <0.1
## supervis   Binary   0.1316 Not Balanced, >0.1
## other      Binary  -0.2766 Not Balanced, >0.1
## ra         Binary   0.1697 Not Balanced, >0.1
## cidi       Binary  -0.3032 Not Balanced, >0.1
## cgneed     Binary  -0.1260 Not Balanced, >0.1
## PSH17A     Binary  -0.0774     Balanced, <0.1
## CRA47A     Binary  -0.0842     Balanced, <0.1
##
##  - - - Subclass 3 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0711
## married    Binary  -0.0912     Balanced, <0.1
## high       Binary   0.1741 Not Balanced, >0.1
## bahigh     Binary  -0.0428     Balanced, <0.1
## poverty2   Binary  -0.1296 Not Balanced, >0.1
## poverty3   Binary   0.0712     Balanced, <0.1
## poverty4   Binary  -0.0406     Balanced, <0.1
## poverty5   Binary   0.1069 Not Balanced, >0.1
## employ     Binary   0.3733 Not Balanced, >0.1
## open       Binary   0.0612     Balanced, <0.1
## black      Binary   0.0648     Balanced, <0.1
## hispanic   Binary  -0.0171     Balanced, <0.1
## natam      Binary  -0.0406     Balanced, <0.1
## cgrage1    Binary   0.1949 Not Balanced, >0.1
## cgrage2    Binary  -0.1040 Not Balanced, >0.1
## cgrage3    Binary  -0.1281 Not Balanced, >0.1
## chdage1    Binary   0.2634 Not Balanced, >0.1
## chdage2    Binary  -0.0534     Balanced, <0.1
## chdage3    Binary  -0.1751 Not Balanced, >0.1
## mental     Binary   0.1614 Not Balanced, >0.1
## arrest     Binary  -0.1188 Not Balanced, >0.1
## sexual     Binary   0.0178     Balanced, <0.1
## provide    Binary  -0.0864     Balanced, <0.1
## supervis   Binary  -0.1823 Not Balanced, >0.1
## other      Binary   0.3035 Not Balanced, >0.1
## ra         Binary   0.0828     Balanced, <0.1
## cidi       Binary   0.0943     Balanced, <0.1
## cgneed     Binary  -0.2083 Not Balanced, >0.1
## PSH17A     Binary   0.0871     Balanced, <0.1
## CRA47A     Binary  -0.2585 Not Balanced, >0.1
##
##  - - - Subclass 4 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0667
## married    Binary   0.3526 Not Balanced, >0.1
## high       Binary  -0.0535     Balanced, <0.1
## bahigh     Binary   0.1308 Not Balanced, >0.1
## poverty2   Binary  -0.3181 Not Balanced, >0.1
## poverty3   Binary   0.2267 Not Balanced, >0.1
```

```
## poverty4   Binary  -0.0258     Balanced, <0.1
## poverty5   Binary   0.2890 Not Balanced, >0.1
## employ     Binary  -0.1127 Not Balanced, >0.1
## open       Binary  -0.2483 Not Balanced, >0.1
## black      Binary  -0.4457 Not Balanced, >0.1
## hispanic   Binary   0.1581 Not Balanced, >0.1
## natam      Binary   0.2906 Not Balanced, >0.1
## cgrage1    Binary   0.0200     Balanced, <0.1
## cgrage2    Binary   0.1367 Not Balanced, >0.1
## cgrage3    Binary  -0.2297 Not Balanced, >0.1
## chdage1    Binary  -0.4153 Not Balanced, >0.1
## chdage2    Binary   0.2267 Not Balanced, >0.1
## chdage3    Binary   0.3390 Not Balanced, >0.1
## mental     Binary  -0.0228     Balanced, <0.1
## arrest     Binary  -0.1273 Not Balanced, >0.1
## sexual     Binary   0.0000     Balanced, <0.1
## provide    Binary  -0.2082 Not Balanced, >0.1
## supervis   Binary  -0.3149 Not Balanced, >0.1
## other      Binary   0.2483 Not Balanced, >0.1
## ra         Binary  -0.0527     Balanced, <0.1
## cidi       Binary   0.5882 Not Balanced, >0.1
## cgneed     Binary   0.2013 Not Balanced, >0.1
## PSH17A     Binary  -0.1464 Not Balanced, >0.1
## CRA47A     Binary   0.1109 Not Balanced, >0.1
##
##  - - - Subclass 5 - - -
##              Type Diff.Adj        M.Threshold
## distance Distance   0.0396
## married    Binary  -0.0120     Balanced, <0.1
## high       Binary  -0.6048 Not Balanced, >0.1
## bahigh     Binary   0.2936 Not Balanced, >0.1
## poverty2   Binary   0.4576 Not Balanced, >0.1
## poverty3   Binary  -0.1908 Not Balanced, >0.1
## poverty4   Binary  -0.2914 Not Balanced, >0.1
## poverty5   Binary  -0.4427 Not Balanced, >0.1
## employ     Binary   0.2115 Not Balanced, >0.1
## open       Binary   0.2688 Not Balanced, >0.1
## black      Binary  -0.1501 Not Balanced, >0.1
## hispanic   Binary   0.2863 Not Balanced, >0.1
## natam      Binary  -0.2914 Not Balanced, >0.1
## cgrage1    Binary  -0.1159 Not Balanced, >0.1
## cgrage2    Binary   0.0832     Balanced, <0.1
## cgrage3    Binary  -0.0050     Balanced, <0.1
## chdage1    Binary  -0.1129 Not Balanced, >0.1
## chdage2    Binary   0.1794 Not Balanced, >0.1
## chdage3    Binary  -0.1095 Not Balanced, >0.1
## mental     Binary   0.4278 Not Balanced, >0.1
## arrest     Binary   0.1393 Not Balanced, >0.1
## sexual     Binary  -0.3729 Not Balanced, >0.1
## provide    Binary   0.3677 Not Balanced, >0.1
## supervis   Binary   0.0363     Balanced, <0.1
## other      Binary  -0.0074     Balanced, <0.1
## ra         Binary  -0.1032 Not Balanced, >0.1
## cidi       Binary  -0.1267 Not Balanced, >0.1
```

```
## cgneed     Binary   0.1389 Not Balanced, >0.1
## PSH17A     Binary  -0.3278 Not Balanced, >0.1
## CRA47A     Binary   0.5694 Not Balanced, >0.1
```

```r
broom::tidy(survival::coxph(f2, df.trimmed, mm2$weights, robust = T))
```

```
## # A tibble: 17 x 6
##    term     estimate std.error robust.se statistic p.value
##    <chr>       <dbl>     <dbl>     <dbl>     <dbl>   <dbl>
##  1 aodserv    0.281      0.160     0.206    1.36    0.173
##  2 married    0.0952     0.193     0.265    0.359   0.720
##  3 high       0.116      0.174     0.218    0.531   0.595
##  4 bahigh    -0.212      0.279     0.304   -0.697   0.486
##  5 poverty2   0.185      0.187     0.239    0.774   0.439
##  6 poverty3  -0.776      0.331     0.355   -2.18    0.0291
##  7 poverty4  -0.132      0.306     0.423   -0.311   0.756
##  8 poverty5  -0.596      0.320     0.506   -1.18    0.239
##  9 employ    -0.0471     0.175     0.239   -0.197   0.844
## 10 open      -0.140      0.194     0.298   -0.469   0.639
## 11 black     -0.558      0.197     0.253   -2.21    0.0272
## 12 hispanic  -0.0485     0.246     0.290   -0.167   0.867
## 13 natam      0.0210     0.265     0.473    0.0444  0.965
## 14 chdage1   -0.399      0.231     0.280   -1.43    0.154
## 15 chdage2    0.118      0.274     0.314    0.377   0.706
## 16 chdage3   -0.259      0.269     0.289   -0.895   0.371
## 17 CRA47A     0.329      0.160     0.204    1.61    0.107
```

```r
exp(0.281)
```

```
## [1] 1.324454
```

Results with Subclassification (MatchIt) and Trimming: The study finds that by controlling for selection bias, children whose caregivers used substance abuse treatment are more likely to receive a maltreatment re-report, and the hazard of having a re-report for these children is 32.45% higher than that of children whose caregivers did not use substance abuse treatment, although this difference is not statistically significant at the .05 level ($p = 0.173$).

Section 5.7: Computer Lab: Running Propensity Score Dosage Analysis with R

Modeling Doses of Treatment With Multiple Balancing Scores Estimated by a Multinomial Logit Model

In this example, the treatment dosage is the categorized percentage of time a child used Aid to Families with Dependent Children (AFDC) from birth to current age in 1997 (`tx0`, `tx1`, and `tx2`). The outcome variable is academic achievement. as measured by the letter-word identification score in 1997 (`lwss97`). We hypothesize that an increase in levels of using AFDC decreases academic achievement in a linear fashion.

Load Packages

We need the `nnet` package to run a multinomial logit model.

```
pacman::p_load(nnet, lmtest, sandwich)
```

Load Data

```
# Load Data
df <- haven::read_dta(here("data", "adv", "chpt10.dta"))

# Function to replicate Stata tab
tab <- function(df, var) {
  df %>%
    group_by({{ var }}) %>%
    summarise(n = n(), .groups = "drop") %>%
    mutate(
      Percent = 100 * round((n / sum(n)), 4),
      Cum. = 100 * round(cumsum(n / sum(n)), 4)
    ) %>%
    rename(Freq. = n)
}
```

Create Treatment Variable

After examining the distribution of percentages of time using AFDC from birth to current age, we create three dose groups: never used, 1% to 33.9% of the time using AFDC from birth to current age in 1997, and 34% to 100% of the time using AFDC from birth to current age in 1997.

In this example, we treated the highest dosage category (i.e., 34% to 100% of the time using AFDC from birth to current age in 1997) as the reference group:

```
tab(df, k_adc)
```

```
## # A tibble: 51 x 4
##    k_adc Freq. Percent  Cum.
##    <dbl> <int>   <dbl> <dbl>
##  1  0      729   72.7   72.7
##  2  7.69    25    2.49  75.2
##  3  8.33    10    1     76.2
##  4  9.09     5    0.5   76.7
##  5 10        4    0.4   77.1
##  6 11.1      8    0.8   77.9
##  7 12.5      1    0.1   78.0
##  8 14.3      1    0.1   78.1
```

```
##  9 15.4     15    1.5   79.6
## 10 16.7      4    0.4   80.0
## # i 41 more rows
```

```r
# Categorize the Continuous Treatment k_adc
df <- df %>%
  mutate(tx = factor(
    case_when(
      k_adc == 0 ~ 0,
      k_adc > 0 & k_adc < 34 ~ 1,
      k_adc >= 34 ~ 2
    ), levels = c(2, 0, 1)))

# Inspect factor levels of treatment
levels(df$tx)
```

```
## [1] "2" "0" "1"
```

The distribution of the three doses are shown below:

```r
tab(df, tx)
```

```
## # A tibble: 3 x 4
##   tx    Freq. Percent  Cum.
##   <fct> <int>   <dbl> <dbl>
## 1 2       139    13.9  13.9
## 2 0       729    72.7  86.5
## 3 1       135    13.5 100
```

Generalized Propensity Scores

Mulitinomial Logit Model To investigate the impact of doses on academic achievement, we employ Imbens's (2000) method. A multinomial logit model was first estimated and the predicted probabilities obtained via the `fitted()` function. Variables showing statistically significant differences from bivariate analyses were entered into the multinomial model as predictors.

We obtain three scores for each child, and each score indicates the generalized propensity of never using AFDC, the generalized propensity of using AFDC for 1% to 33.9% of the time, and the generalized propensity of using AFDC for 34% to 100% of the time.

```r
# Use multinomial logit model to predict generalized propensity scores
m1 <- nnet::multinom(
  tx ~ pcg_adc + age97 + mratio96 + pcged97 + black,
  data = df
)
```

```
## # weights:  21 (12 variable)
## initial  value 1101.908126
## iter  10 value 651.499728
## iter  20 value 534.660846
## final  value 534.657501
## converged
```

```r
summary(m1)
```

```
## Call:
## nnet::multinom(formula = tx ~ pcg_adc + age97 + mratio96 + pcged97 +
##     black, data = df)
##
```

```
## Coefficients:
##   (Intercept)    pcg_adc       age97 mratio96    pcged97      black
## 0  -2.6933481 -0.2758773 -0.11732695 1.767833 0.29748922 -0.9878301
## 1  -0.4118176 -0.1550842 -0.08486823 1.032242 0.03898852 -0.2768833
##
## Std. Errors:
##   (Intercept)    pcg_adc      age97  mratio96    pcged97     black
## 0   0.9788544 0.05545970 0.04258354 0.1859115 0.07719240 0.3033386
## 1   1.0171423 0.05356707 0.04483509 0.1935113 0.08088266 0.3349232
##
## Residual Deviance: 1069.315
## AIC: 1093.315
```

```r
# Save model predicted probabilities
head(fitted(m1))
```

```
##             2         0          1
## 1 0.146603075 0.6463734 0.20702356
## 2 0.495585081 0.2469185 0.25749645
## 3 0.000244027 0.9835595 0.01619643
## 4 0.175124578 0.5883837 0.23649169
## 5 0.146015065 0.6203262 0.23365875
## 6 0.009837973 0.9057460 0.08441600
```

```r
pr <- fitted(m1) %>%
  as_tibble() %>%
  setNames(c("pr2", "pr0", "pr1"))
head(pr)
```

```
## # A tibble: 6 x 3
##        pr2   pr0    pr1
##      <dbl> <dbl>  <dbl>
## 1 0.147    0.646 0.207
## 2 0.496    0.247 0.257
## 3 0.000244 0.984 0.0162
## 4 0.175    0.588 0.236
## 5 0.146    0.620 0.234
## 6 0.00984  0.906 0.0844
```

```r
summary(pr)
```

```
##       pr2                pr0               pr1
##  Min.   :0.000000   Min.   :0.02348   Min.   :0.00000
##  1st Qu.:0.002256   1st Qu.:0.52566   1st Qu.:0.03719
##  Median :0.029368   Median :0.84854   Median :0.12447
##  Mean   :0.138588   Mean   :0.72682   Mean   :0.13460
##  3rd Qu.:0.201214   3rd Qu.:0.96053   3rd Qu.:0.22369
##  Max.   :0.885824   Max.   :1.00000   Max.   :0.42421
```

Sampling Weights

Note that even though each participant has three generalized propensity scores (GPS), the GPS for the dose category each participant actually used is chosen for the weighted analysis. Below is the key syntax to create the weight variable (`dose_w`). It is important to note that the GPS itself is not the weight; the inverse of a specific GPS is defined as a weight.

```r
# Define inverse of the generalized p-scores as sampling weights
df <- df %>%
  mutate(dose_w = case_when(
    tx == 0 ~ 1 / pr$pr0,
    tx == 1 ~ 1 / pr$pr1,
    tx == 2 ~ 1 / pr$pr2
  ))
summary(df$dose_w)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##    1.000    1.043    1.208    6.648    2.206 2611.475
```

Dummy Variables

Next, we create dummy variables for the outcome analysis. For a study with three dosage categories, we create $d - 1 = 3 - 1 = 2$ dummy variables. These variables then are entered into the outcome analysis. In this example, the outcome analysis is a weighted regression.

```r
# Create t-1 dummy variables to measure dose categories
df <- df %>%
  mutate(
    tx0 = ifelse(tx == 0, 1, 0),
    tx1 = ifelse(tx == 1, 1, 0)
  )

tab(df, tx0)
```

```
## # A tibble: 2 x 4
##     tx0 Freq. Percent  Cum.
##   <dbl> <int>   <dbl> <dbl>
## 1     0   274    27.3  27.3
## 2     1   729    72.7 100
```

```r
tab(df, tx1)
```

```
## # A tibble: 2 x 4
##     tx1 Freq. Percent  Cum.
##   <dbl> <int>   <dbl> <dbl>
## 1     0   868    86.5  86.5
## 2     1   135    13.5 100
```

Regression Analysis

With Propensity Score Adjustment As a reminder, note that `tx0` is never used AFDC, `tx1` is using AFDC 1% to 33.9%, and `tx2` is using AFDC 34% to 100% of the time from birth to current age in 1997.

The results of the regression with propensity score adjustments show that, other things being equal, (a) "using AFDC 1% to 33.9% of the time" decreases the academic achievement from "never used" by 4.19 units (i.e., $7.48 - 3.29 = 4.19$), (b) "using AFDC 34% to 100% of the time" decreases the academic achievement from "using AFDC 1% to 33.9% of the time" by 3.29 units, and (c) "using AFDC 34% to 100% of the time" decreases the academic achievement from "never used" by 7.48 units. All differences are statistically significant at a .05 level from a one-tailed test. Thus, the results confirm a linear impact of doses of using AFDC on academic achievement: The longer the time a study child was exposed to poverty, the lower his or her academic achievement. Such a conclusion cannot be drawn by analyses treating the child's use of AFDC as a binary variable.

```r
# Regression controls for selection using inverse p-scores as weights
m3 <- lm(lwss97 ~ male + black + age97 + pcged97 + mratio96 + pcg_adc + tx0 + tx1,
         weights = dose_w, data = df)
lmtest::coeftest(m3, vcovCL, cluster = df$pcg_id)
```

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value              Pr(>|t|)
## (Intercept) 73.32126    3.99587 18.3493 < 0.00000000000000022 ***
## male        -1.17070    1.30288 -0.8985             0.3691148
## black       -1.58184    1.95036 -0.8110             0.4175313
## age97        0.65806    0.20802  3.1635             0.0016062 **
## pcged97      1.25721    0.37837  3.3227             0.0009240 ***
## mratio96     1.02647    0.28148  3.6467             0.0002794 ***
## pcg_adc     -0.48280    0.53557 -0.9015             0.3675611
## tx0          7.48356    1.06752  7.0102     0.000000000004388 ***
## tx1          3.28909    1.73504  1.8957             0.0582914 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Without Propensity Score Adjustment In contrast, the regression model without propensity score adjustments showed the impact of poverty in a lesser amount for all possible contrasts, and only one of the two treatment variables was statistically significant. This exemplifies, in a sense, the importance of controlling for covariates using propensity scores. Together, the analyses show that modeling multiple doses of treatment is informative. It allows us to test research hypotheses not only centering on treatment effectiveness but also concerning the impact of dose levels.

```r
# Regression does not control for selection
m2 <- lm(lwss97 ~ male + black + age97 + pcged97 + mratio96 + pcg_adc + tx0 +
  tx1, data = df)
lmtest::coeftest(m2, vcovCL, cluster = df$pcg_id)
```

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value              Pr(>|t|)
## (Intercept) 79.40538    4.44405 17.8678 < 0.00000000000000022 ***
## male        -2.01599    0.98653 -2.0435             0.0412657 *
## black       -1.85547    1.23201 -1.5061             0.1323713
## age97        0.87847    0.16895  5.1995          0.0000002425 ***
## pcged97      0.90784    0.33594  2.7024             0.0070013 **
## mratio96     1.12921    0.33211  3.4001             0.0007003 ***
## pcg_adc     -0.72704    0.31807 -2.2858             0.0224783 *
## tx0          5.44600    1.79756  3.0297             0.0025116 **
## tx1          1.29861    1.97936  0.6561             0.5119295
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Hirano and Imbens Estimator

Load Package

The package `causaldrf` implements the Hirano and Imbens estimator in R, but some of the arguments are different from those in Stata and therefore yield different results.

```
pacman::p_load(causaldrf)
```

Manage Data

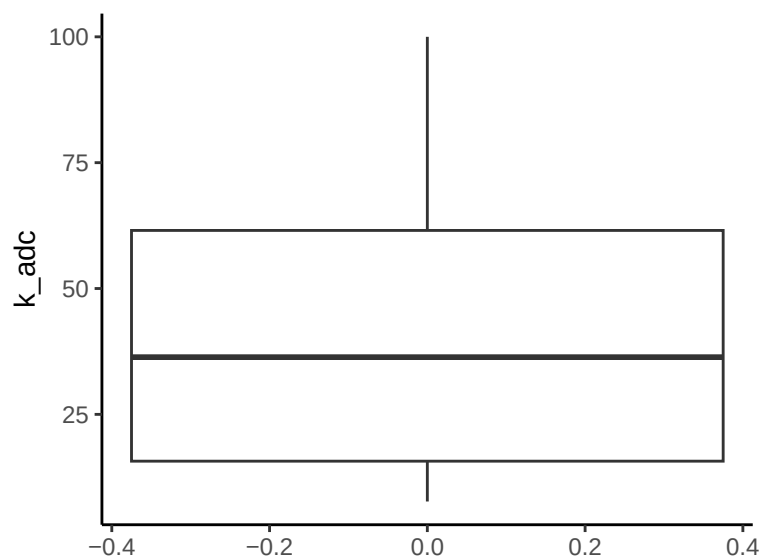
For this example, we will be using the same data and research question from above, but instead treat the treatment dosage as a continuous variable, the percentage of time a child used AFDC from birth to 1997.

We first check the normality of the treatment dosage. If it is highly skewed, we should transform the treatment dosage in the following steps.

We first excluded all participants with a zero dosage (i.e., `k_adc == 0`). We then divided the sample into three groups based on the distribution of treatment dosage: [1, 25], (25, 60], and (60, 100]. The three groups are those who used the program from birth to 1997 for 1% to 25% of the time, 25% to 60% of the time, and 60% to 100% of the time. The sample size for each group is 103, 101, and 70, respectively. There is no specific guidance on how to choose cutoff points and the number of k in this process. In practice, it makes sense to divide the entire sample into a few equally sized groups.

```
# Load Data
df <- haven::read_dta(here("data", "adv", "chpt10.dta"))

# Treatment distribution
df %>%
  filter(k_adc != 0) %>%
  ggplot(aes(y = k_adc)) +
  geom_boxplot() +
  theme_classic()
```




```
# Check normality of the treatment dosage using the Kolmogorov-Smirnov test
# Not normal, therefore transformation needed
ks.test(df$k_adc, "pnorm")
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
```

```
##
## data:  df$k_adc
## D = 0.5, p-value < 0.00000000000000022
## alternative hypothesis: two-sided
```

```r
# Manipulate Data
df2 <- df %>%

  # Exclude participants with zero dosage
  filter(k_adc != 0) %>%

  # Divide sample into three groups based on distribution of treatment dosage
  mutate(cut = ifelse(k_adc <= 25, 25, # [1, 25]
               ifelse(k_adc > 25 & k_adc <= 60, 60, # (25, 60]
               ifelse(k_adc > 60, 100, NA)))) %>% # (60, 100]

  # Rename variables
  rename(
    treat = k_adc,
    Y = lwss97
  )

# Sample Size for Each Group
df2 %>% count(cut)
```

```
## # A tibble: 3 x 2
##     cut     n
##   <dbl> <int>
## 1    25   103
## 2    60   101
## 3   100    70
```

```r
# Calculate the median treatment dosage of each group
grid_val_values <- df2 %>%
  group_by(cut) %>%
  summarize(treat_median = median(treat)) %>%
  pull(treat_median)
grid_val_values
```

```
## [1] 11.11111 40.00000 75.00000
```

Estimate Median Treatment of Each Group

The function `hi_est()` estimates the GPS function and the ADRF. The Hirano-Imbens estimator requires two models. The `treat_formula` regresses the treatment on the set of covariates to estimate the GPS values. The `outcome_formula` regresses the outcome on the observed treatment and fitted GPS values. The output provides the fit of both the treatment and outcome model, and the estimated outcome values `param` on the grid of treatment values in `grid_val`. We can specify the error distribution to be used in the model for treatment with `treat_mod`.

If the F statistic for the outcome model is significant, one concludes that the covariates introduce bias.

```r
hi_list0 <- causaldrf::hi_est(
  Y = Y,
  treat = treat,
  treat_formula = treat ~ pcg_adc + age97 + male + black + mratio96 + pcged97,
  outcome_formula = Y ~ treat + I(treat^2) + gps + I(gps^2) + treat * gps,
```

```r
  data = df2,
  grid_val = grid_val_values,
  treat_mod = "LogNormal"
)
hi_sum0 <- summary(hi_list0)
```

```
##
## Estimated values:
## [1] 95.46135 93.24527 92.25934
##
## Treatment Summary:
##
## Call:
## lm(formula = formula_t, data = samp_dat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9321 -0.4831  0.1554  0.5468  1.2850
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  3.908791   0.353262  11.065 < 0.0000000000000002 ***
## pcg_adc      0.042645   0.017374   2.454               0.0147 *
## age97        0.002834   0.015280   0.185               0.8530
## male        -0.083815   0.086600  -0.968               0.3340
## black        0.185549   0.112681   1.647               0.1008
## mratio96    -0.283830   0.045688  -6.212        0.00000000199 ***
## pcged97     -0.033444   0.028680  -1.166               0.2446
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7083 on 267 degrees of freedom
## Multiple R-squared:  0.2101, Adjusted R-squared:  0.1923
## F-statistic: 11.83 on 6 and 267 DF,  p-value: 0.000000000009066
##
##
## Outcome Summary:
##
## Call:
## lm(formula = outcome_formula, data = tempdat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -38.010  -8.782  -0.375   8.283  53.476
##
## Coefficients:
##               Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  91.258266   4.901010  18.620 <0.0000000000000002 ***
## treat        -0.121650   0.227484  -0.535               0.593
## I(treat^2)    0.001650   0.001877   0.879               0.380
## gps          38.355981  31.178846   1.230               0.220
## I(gps^2)    -32.571974  48.154665  -0.676               0.499
## treat:gps    -0.326382   0.330943  -0.986               0.325
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.78 on 268 degrees of freedom
## Multiple R-squared:  0.02465,    Adjusted R-squared:  0.006454
## F-statistic: 1.355 on 5 and 268 DF,  p-value: 0.2418
```

```r
hi_list0$param # estimated outcome values
```

```
## [1] 95.46135 93.24527 92.25934
```

Estimate the Average Dose Response Function (ADRF)

We now move to the last stage to estimate the dose response function and treatment effect function (i.e., the estimated derivative). Note: we specified a 10-unit increment of the treatment dosage for the x-axis in the plot using the `grid_val()` argument.

```r
hi_list <- causaldrf::hi_est(
  Y = Y,
  treat = treat,
  treat_formula = treat ~ pcg_adc + age97 + male + black + mratio96 + pcged97,
  outcome_formula = Y ~ treat + I(treat^2) + gps + I(gps^2) + treat * gps,
  data = df2,
  grid_val = seq(10, 100, by = 10),
  treat_mod = "LogNormal"
)
hi_sum <- summary(hi_list)
```

```
##
## Estimated values:
##  [1] 94.89641 96.81063 95.01335 93.24527 92.20739 91.82329 91.98884 92.64925
##  [9] 93.77512 95.34519
##
## Treatment Summary:
##
## Call:
## lm(formula = formula_t, data = samp_dat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9321 -0.4831  0.1554  0.5468  1.2850
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  3.908791   0.353262  11.065 < 0.0000000000000002 ***
## pcg_adc      0.042645   0.017374   2.454               0.0147 *
## age97        0.002834   0.015280   0.185               0.8530
## male        -0.083815   0.086600  -0.968               0.3340
## black        0.185549   0.112681   1.647               0.1008
## mratio96    -0.283830   0.045688  -6.212        0.00000000199 ***
## pcged97     -0.033444   0.028680  -1.166               0.2446
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7083 on 267 degrees of freedom
## Multiple R-squared:  0.2101, Adjusted R-squared:  0.1923
## F-statistic: 11.83 on 6 and 267 DF,  p-value: 0.000000000009066
```

```
##
##
## Outcome Summary:
##
## Call:
## lm(formula = outcome_formula, data = tempdat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -38.010  -8.782  -0.375   8.283  53.476
##
## Coefficients:
##               Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  91.258266   4.901010  18.620 <0.0000000000000002 ***
## treat        -0.121650   0.227484  -0.535               0.593
## I(treat^2)    0.001650   0.001877   0.879               0.380
## gps          38.355981  31.178846   1.230               0.220
## I(gps^2)    -32.571974  48.154665  -0.676               0.499
## treat:gps    -0.326382   0.330943  -0.986               0.325
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.78 on 268 degrees of freedom
## Multiple R-squared:  0.02465,    Adjusted R-squared:  0.006454
## F-statistic: 1.355 on 5 and 268 DF,  p-value: 0.2418
```

```r
hi_list$param
```

```
##  [1] 94.89641 96.81063 95.01335 93.24527 92.20739 91.82329 91.98884 92.64925
##  [9] 93.77512 95.34519
```

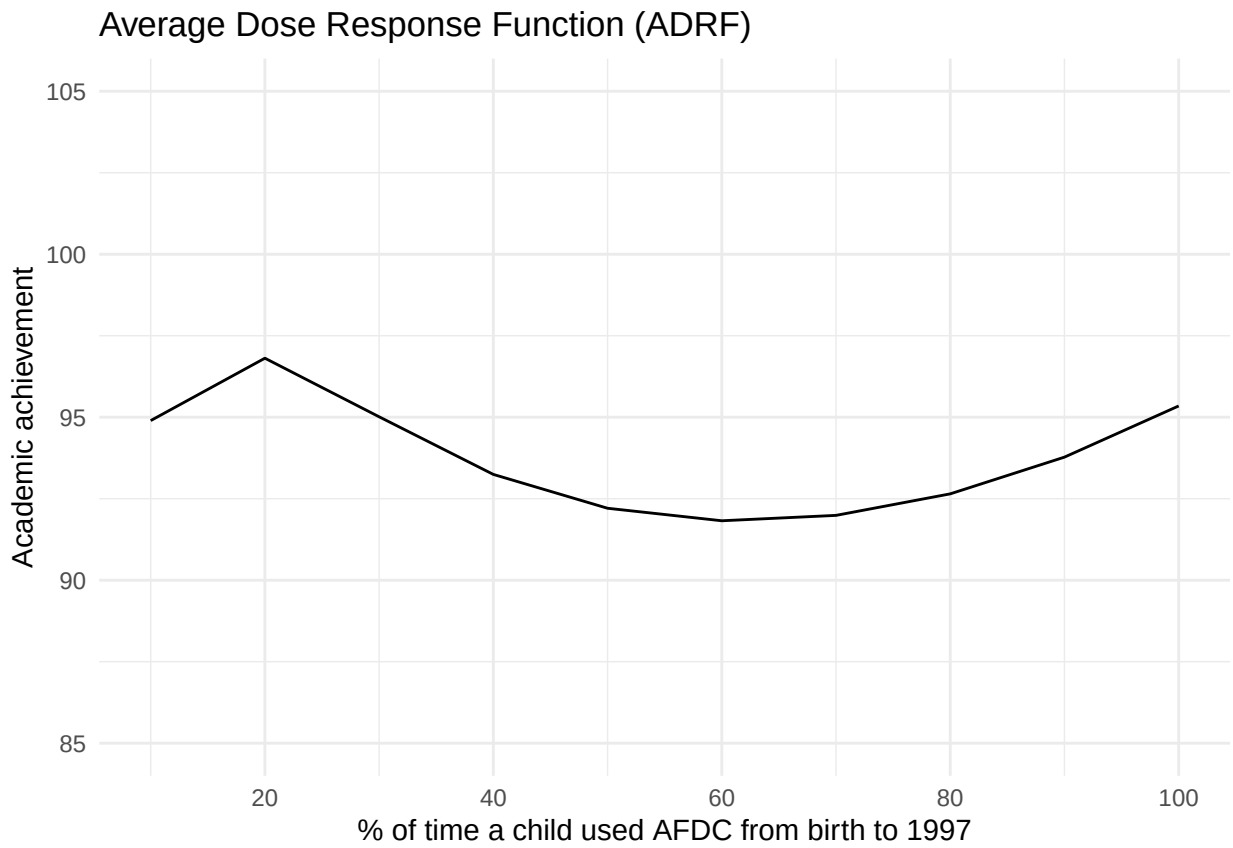
Plot ADRF

The dose-response functions show that children who used the welfare program 10% to 30% of the time from birth to 1997 have a letter-word identification score around 95; after the 30% threshold, an increase in using the welfare program decreases the letter-word identification score, and the lowest achievement score is linked to children who used the welfare program 60% of the time; after the 60% threshold, the letter-word identification score gradually increases with every unit increase in the level of using the welfare program. The achievement for children who used AFDC 100% of the time recovers to the level around 95 or, more precisely, to the level of 96.

The results suggest that for children with high AFDC participation (i.e., more consistently in poverty during childhood), academic achievement stabilizes at a higher level compared to children who used the welfare program about 60% of the time (i.e., were inconsistently in poverty). The findings may suggest that children who were stably on AFDC did better than children who experienced the trauma of cycling into and out of poverty. In other words, given the findings, one may wonder if the trauma of going in and out of AFDC implies a more chaotic childhood.

```r
adrf <- tibble(
  x = seq(10, 100, by = 10),
  y = hi_list$param
)
ggplot(adrf, aes(x = x, y = y)) +
  geom_line() +
  theme_minimal() +
  ylim(85, 105) +
```

```r
  scale_x_continuous(breaks = seq(0, 100, by = 20)) +
  labs(
    title = "Average Dose Response Function (ADRF)",
    x = "% of time a child used AFDC from birth to 1997",
    y = "Academic achievement"
  )
```



Balance Check

Below we run unadjusted t-tests for each covariate:

```r
# Test Unadjusted mean differences
# Dummy code the treatment groups
df3 <- df2 %>%
  mutate(g1 = ifelse(treat <= 25, 1, 0),
         g2 = ifelse(treat > 25 & treat <= 60, 1, 0),
         g3 = ifelse(treat > 60, 1, 0))
df3 %>% count(cut, g1)
```

```
## # A tibble: 3 x 3
##     cut    g1     n
##   <dbl> <dbl> <int>
## 1    25     1   103
## 2    60     0   101
## 3   100     0    70
```

```r
df3 %>% count(cut, g2)
```

```
## # A tibble: 3 x 3
##     cut    g2     n
##   <dbl> <dbl> <int>
## 1    25     0   103
## 2    60     1   101
## 3   100     0    70
```

```r
df3 %>% count(cut, g3)
```

```
## # A tibble: 3 x 3
##     cut    g3     n
##   <dbl> <dbl> <int>
## 1    25     0   103
## 2    60     0   101
## 3   100     1    70
```

```r
# Compare means of each covariate in each treatment group
unadjust <- function(cov, group) {
  cov_v <- df3 %>% select(all_of(cov)) %>% pull()
  group_v <- df3 %>% select(all_of(group)) %>% pull()
  t.test(cov_v ~ group_v, var.equal = T) %>%
    broom::tidy() %>%
    mutate(Formula = paste0(cov, " ~ ", group)) %>%
    select(Formula, MeanGroup0 = estimate1, MeanGroup1 = estimate2,
           t = statistic, p.value, df = parameter)
}
arg1 <- rep(c("pcg_adc", "age97", "male", "black", "mratio96", "pcged97"), each = 3)
arg2 <- rep(c("g1", "g2", "g3"), 6)
map2_dfr(arg1, arg2, unadjust)
```

```
## # A tibble: 18 x 6
##    Formula       MeanGroup0 MeanGroup1      t      p.value    df
##    <chr>              <dbl>      <dbl>  <dbl>        <dbl> <dbl>
##  1 pcg_adc ~ g1       2.14       1.41   2.29  0.0229         272
##  2 pcg_adc ~ g2       2.06       1.52   1.67  0.0964         272
##  3 pcg_adc ~ g3       1.47       3.03  -4.51  0.00000956     272
##  4 age97 ~ g1         7.16       7.05   0.311 0.756          272
##  5 age97 ~ g2         7.24       6.90   0.968 0.334          272
##  6 age97 ~ g3         6.98       7.53  -1.42  0.157          272
##  7 male ~ g1          0.515      0.524 -0.154 0.877          272
##  8 male ~ g2          0.486      0.574 -1.42  0.157          272
##  9 male ~ g3          0.549      0.429  1.74  0.0824         272
## 10 black ~ g1         0.842      0.680  3.20  0.00155        272
## 11 black ~ g2         0.769      0.802 -0.639 0.523          272
## 12 black ~ g3         0.740      0.9   -2.82  0.00516        272
## 13 mratio96 ~ g1      0.794      1.48  -5.80  0.0000000182   272
## 14 mratio96 ~ g2      1.13       0.924  1.61  0.109          272
## 15 mratio96 ~ g3      1.20       0.605  4.46  0.0000119      272
## 16 pcged97 ~ g1      11.5       11.8   -1.51  0.132          272
## 17 pcged97 ~ g2      11.5       11.8   -1.70  0.0911         272
## 18 pcged97 ~ g3      11.8       11.0    3.62  0.000351       272
```

Post-adjustment balance checks can be conducted by following the procedure in Galagate & Schafer (2015).[7]

[7] https://rdrr.io/cran/causaldrf/f/inst/doc/Using_causaldrf.pdf

Addendum

This section presents additional R packages that estimate causal effects for multinomial or continuous treatments.

```
pacman::p_load(twangContinuous, WeightIt, survey, dbarts, mlogit)
```

twangContinuous

The `twangContinuous` package can estimate causal effects for continuous treatments using generalized propensity scores estimated via generalized boosted models. For example, we may want to estimate the effect of trauma symptoms (`tss_0`) on substance use frequency among individuals with substance use disorder (`sfs8p_3`).[8]

```
# Load sample data
data(dat)

# Set seed and estimate GPS with GBM
set.seed(1234)
test.mod <- twangContinuous::ps.cont(
  tss_0 ~ sfs8p_0 + sati_0 + sp_sm_0
    + recov_0 + subsgrps_n + treat,
  data = dat,
  n.trees = 500,
  shrinkage = 0.01,
  interaction.depth = 3,
  verbose = FALSE
)

# Check balance to see if correlation between treatment and covariate has reduced
# unw = unweighted correlation between treatment and covariate
# wcor = weighted correlation between treatment and covariate
# Below .1 is a good threshold
twangContinuous::bal.table(test.mod, digits = 2)
```

```
##              unw  wcor
## sfs8p_0     0.12 -0.01
## sati_0      0.14 -0.01
## sp_sm_0     0.22  0.00
## recov_0    -0.11 -0.04
## subsgrps_n  0.04 -0.05
## treatA      0.05  0.02
## treatB     -0.05 -0.02
```

```
# Extract propensity score weights
dat$wts <- get.weights(test.mod)

# Outcome analysis
design.ps <- survey::svydesign(ids = ~1, weights = ~wts, data = dat)
outcome.model <- survey::svyglm(sfs8p_3 ~ tss_0,
  design = design.ps,
  family = gaussian()
)

# For each additional trauma symptom, substance use frequency
```

[8] https://cran.r-project.org/web/packages/twangContinuous/vignettes/briefTutorial.html

```r
# increases by .0026, t = .042, p = .97
summary(outcome.model)
```

```
##
## Call:
## svyglm(formula = sfs8p_3 ~ tss_0, design = design.ps, family = gaussian())
##
## Survey design:
## survey::svydesign(ids = ~1, weights = ~wts, data = dat)
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## (Intercept) 6.595461   0.232069  28.420 <0.0000000000000002 ***
## tss_0       0.002616   0.062416   0.042               0.967
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 111.6769)
##
## Number of Fisher Scoring iterations: 2
```

WeightIt

The WeightIt package supports binary, multinomial, and continuous treatments, such as the CBPS method, GBM, and entropy balancing.[9]

```r
# Import data
data("lalonde", package = "cobalt")

# Continuous treatment using energy balancing
(w.cont <- WeightIt::weightit(
  re75 ~ age + educ + race + married +
    nodegree + re74,
  data = lalonde,
  method = "energy"
))
```

Continuous Treatment

```
## A weightit object
##  - method: "energy" (energy balancing)
##  - number of obs.: 614
##  - sampling weights: none
##  - treatment: continuous
##  - covariates: age, educ, race, married, nodegree, re74
```

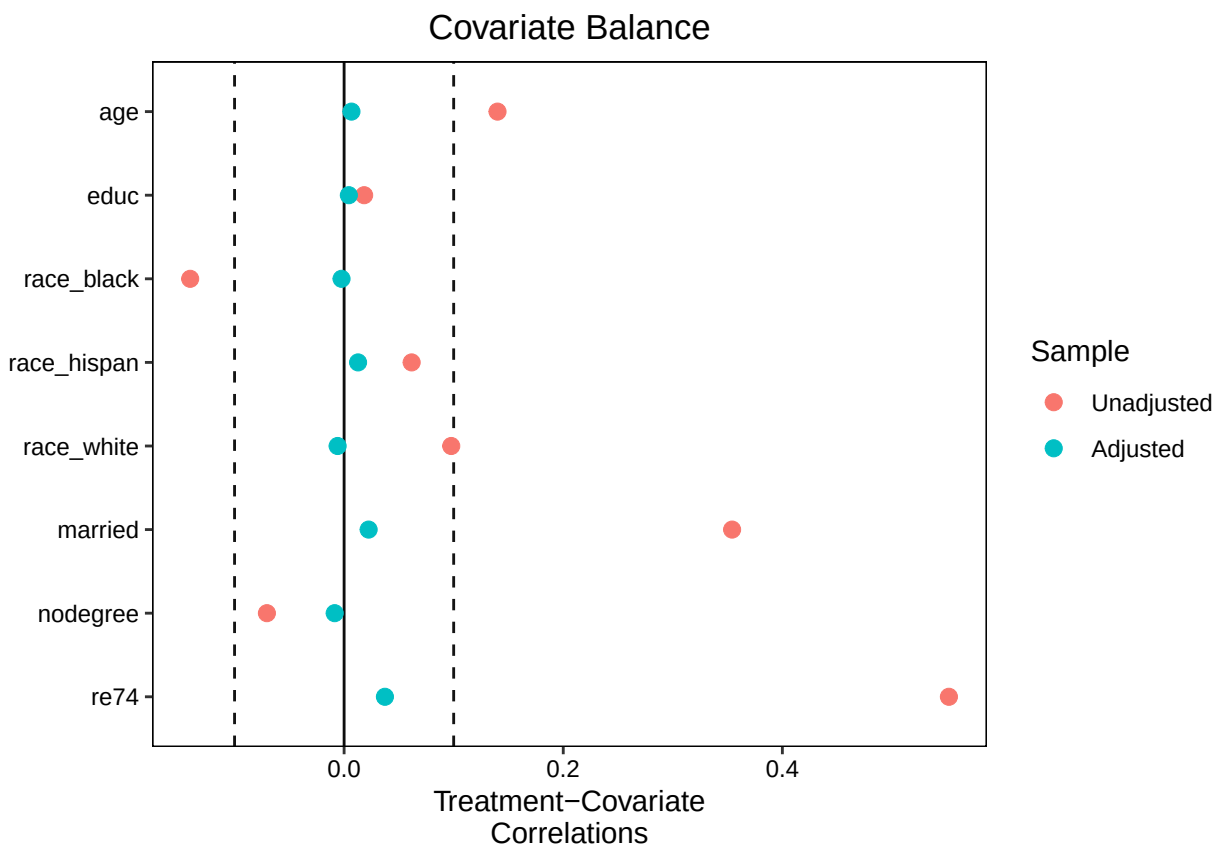
```r
# Check balance by assessing correlations between treatment and covariates
cobalt::bal.tab(w.cont, un = T)
```

```
## Balance Measures
##                Type Corr.Un Corr.Adj
## age         Contin.  0.1400   0.0067
## educ        Contin.  0.0183   0.0043
```

[9] https://cran.r-project.org/web/packages/cobalt/vignettes/cobalt.html#using-cobalt-with-continuous-treatments

```
## race_black   Binary -0.1405  -0.0023
## race_hispan  Binary  0.0616   0.0127
## race_white   Binary  0.0978  -0.0059
## married      Binary  0.3541   0.0224
## nodegree     Binary -0.0705  -0.0086
## re74        Contin.  0.5520   0.0372
##
## Effective sample sizes
##            Total
## Unadjusted 614.
## Adjusted   299.9
```

```
cobalt::love.plot(w.cont, thresholds = c(cor = .1))
```




```
# Weights for outcome analysis
head(w.cont$weights)
```

```
## [1] 0.7265275078664 0.0000001571634 0.3104532119795 0.6619353942574
## [5] 0.5926962365091 0.4547616197067
```

Multinomial Treatment The covariate balancing propensity score (CBPS) can be used for binary, multitreatment, continuous, or longitudinal treatments. For finer control, the `CBPS` package contains more options than the `WeightIt` package.

```
# Define treatment formula
f <- as.formula("race ~ age + educ + married + nodegree + re74")

# Obtain weights via CBPS
(w.multi <- WeightIt::weightit(f, data = lalonde, method = "cbps", estimand = "ATE"))
```

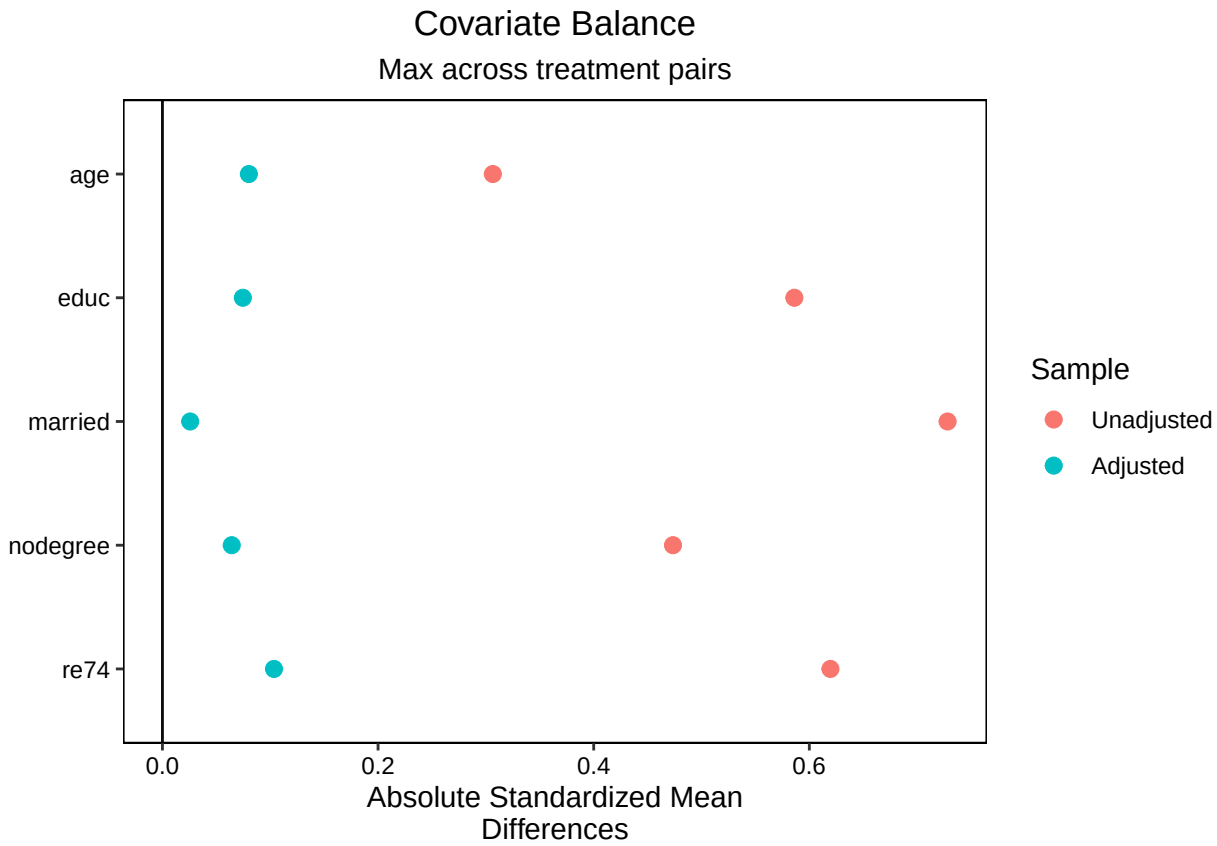
```
## A weightit object
##  - method: "cbps" (covariate balancing propensity score weighting)
##  - number of obs.: 614
##  - sampling weights: none
##  - treatment: 3-category (black, hispan, white)
##  - estimand: ATE
##  - covariates: age, educ, married, nodegree, re74
```

```r
# Balance check
cobalt::bal.tab(w.multi, un = T, which.treat = .all)
```

```
## Balance by treatment pair
##
##  - - - black (0) vs. hispan (1) - - -
## Balance Measures
##             Type Diff.Un Diff.Adj
## age      Contin. -0.0101  -0.0802
## educ     Contin. -0.4515  -0.0738
## married   Binary  0.2222  -0.0121
## nodegree  Binary  0.0684  -0.0163
## re74     Contin.  0.3183   0.0207
##
## Effective sample sizes
##             black hispan
## Unadjusted 243.    72.
## Adjusted   185.35  56.75
##
##  - - - black (0) vs. white (1) - - -
## Balance Measures
##             Type Diff.Un Diff.Adj
## age      Contin.  0.2963  -0.0299
## educ     Contin.  0.1347  -0.0746
## married   Binary  0.3430  -0.0048
## nodegree  Binary -0.1503   0.0134
## re74     Contin.  0.6196   0.1035
##
## Effective sample sizes
##             black  white
## Unadjusted 243.   299.
## Adjusted   185.35 257.77
##
##  - - - hispan (0) vs. white (1) - - -
## Balance Measures
##             Type Diff.Un Diff.Adj
## age      Contin.  0.3065   0.0503
## educ     Contin.  0.5861  -0.0008
## married   Binary  0.1208   0.0073
## nodegree  Binary -0.2187   0.0297
## re74     Contin.  0.3013   0.0828
##
## Effective sample sizes
##            hispan  white
## Unadjusted  72.   299.
```

```
## Adjusted    56.75 257.77
##  - - - - - - - - - - - - - - - - - - - -
```

```r
cobalt::love.plot(w.multi, binary = "std") # which.treat = .all
```



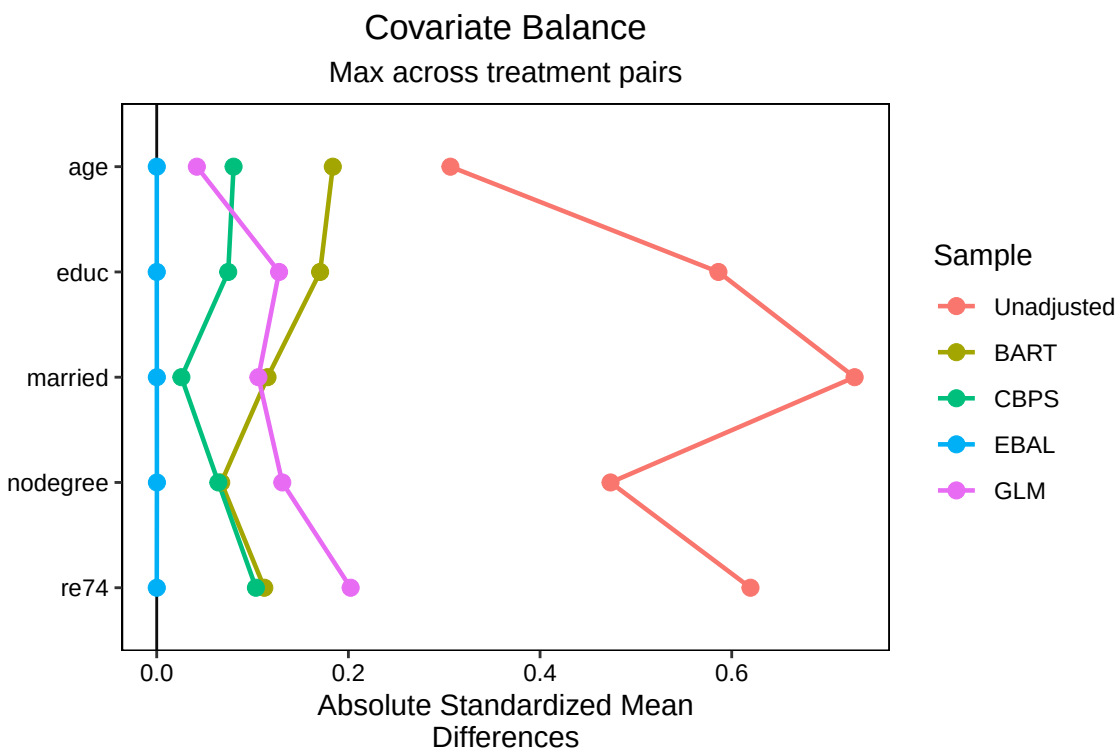

```r
# Weights for outcome analysis
head(w.multi$weights)
```

```
## [1] 3.508239 7.337808 2.036473 1.662737 1.692773 1.722472
```

```r
m1 <- WeightIt::weightit(f, lalonde, "bart", estimand = "ATE")
m2 <- WeightIt::weightit(f, lalonde, "cbps", estimand = "ATE")
m3 <- WeightIt::weightit(f, lalonde, "ebal", estimand = "ATE")
m4 <- WeightIt::weightit(f, lalonde, "glm", estimand = "ATE")
```

```r
cobalt::love.plot(f,
  data = lalonde,
  weights = list(
    BART = m1,
    CBPS = m2,
    EBAL = m3,
    GLM = m4
  ),
  line = T,
  s.d.denom = "pooled",
  #which.treat = .all,
  binary = "std"
)
```

Comparing Methods



Section 6.4: Computer Lab: Running Sensitivity Analysis with rbounds

Sensitivity analysis seeks to answer the question: How sensitive are findings to hidden bias? Although sensitivity analysis is exploratory, it is an important step and recommended in observational studies. In this section, we demonstrate how to carry out Rosenbaum's approach to sensitivity analysis.

Sensitivity Analysis of the Effects

The steps in conducting the sensitivity analysis for a matched pair study are as follows:

1. Arrange the data (i.e., the outcome variable) in pairs, then list the value of outcome variable for the treated and control units separately, and take the difference between the two values at the pair level;
2. Use the Wilcoxon's signed-rank test to calculate the lower and upper bounds of p-values when $\Gamma = 1$;
3. Repeat Step 2 for different values of Γ; that is, calculate the lower and upper bounds of p-values when $\Gamma = 2, 3, \cdots$ until a Γ value at which the p-value is $> .05$; this is the Γ value at which the significance interval becomes uninformative. The Γ value so obtained is the measure of sensitivity to hidden bias. A large Γ is good, meaning that the study is insensitive to hidden selection bias.

Interpretation: If $\Gamma = 2$, then two units that appear similar or have the same covariates could differ in their odds of receiving treatment by as much as a factor of 2, so one unit might be twice as likely as the other to receive treatment. Thus, "Γ is a measure of the degree of departure from a study that is free of hidden bias. A sensitivity analysis will consider several possible values of Γ and show how the inferences might change. A study is sensitive if values of Γ close to 1 could lead to inferences that are very different from those obtained assuming the study is free of hidden bias. A study is insensitive if extreme values of Γ are required to alter the inference" (Rosenbaum, 2002, p. 107).

Load Packages

We will use the packages `rbounds` for the sensitivity analysis and `coin` for the Wilcoxon signed-rank test.

```
pacman::p_load(rbounds, coin, kableExtra, Matching, conflicted)
conflicted::conflicts_prefer(dplyr::select(), dplyr::filter())
```

Load Data

The data below contains blood lead levels (ug/dl) of children whose parents were exposed to lead at their places of work (`trt`), versus children whose parents were not exposed to lead at their places of work (`ctrl`):

```
# Data: Matched Data of Lead Blood Levels in Children
# Source: Rosenbaum (2002a), p. 82
df.sa <- tibble(
  trt = c(
    38, 23, 41, 18, 37, 36, 23, 62, 31, 34, 24, 14, 21, 17, 16, 20, 15, 10, 45,
    39, 22, 35, 49, 48, 44, 35, 43, 39, 34, 13, 73, 25, 27
  ),
  ctrl = c(
    16, 18, 18, 24, 19, 11, 10, 15, 16, 18, 18, 13, 19, 10, 16, 16, 24, 13, 9,
    14, 21, 19, 7, 18, 19, 12, 11, 22, 25, 16, 13, 11, 13
  )
)
head(df.sa)
```

```
## # A tibble: 6 x 2
##     trt  ctrl
##   <dbl> <dbl>
```

```
## 1    38    16
## 2    23    18
## 3    41    18
## 4    18    24
## 5    37    19
## 6    36    11
```

```r
summary(df.sa)
```

```
##       trt             ctrl
##  Min.   :10.00   Min.   : 7.00
##  1st Qu.:21.00   1st Qu.:13.00
##  Median :34.00   Median :16.00
##  Mean   :31.85   Mean   :15.88
##  3rd Qu.:39.00   3rd Qu.:19.00
##  Max.   :73.00   Max.   :25.00
```

Rosenbaum Bounds for Sign Rank

Below, we see the range of significance levels for the test statistic when Γ is equal to various values. As depicted, $\Gamma = 4.25$ is the value at which the significance interval becomes uninformative. Rosenbaum (2002) interpreted this finding in the following way: "The table shows that to explain away the observed association between parental exposure to lead and child's lead level, a hidden bias or unobserved covariate would need to increase the odds of exposure by more than a factor of $\Gamma = 4.25$" (p. 115).

```r
p <- rbounds::psens(df.sa$trt, df.sa$ctrl, GammaInc = 0.25)
p$bounds %>%
  filter(Gamma %in% c(1:4, 4.25, 5:6))
```

```
##   Gamma Lower bound Upper bound
## 1  1.00           0      0.0000
## 2  2.00           0      0.0018
## 3  3.00           0      0.0131
## 4  4.00           0      0.0367
## 5  4.25           0      0.0441
## 6  5.00           0      0.0694
## 7  6.00           0      0.1073
```

Note that there are additional functions available in the `rbounds` package, such as `rbounds::hlsens()`, which will calculate the Rosenbaum bounds for continuous or ordinal outcomes based on the Hodges-Lehmann point estimate.

Example of Sensitivity Analysis & Pair Matching

A study using regression analysis of difference scores found that in 1997, children who used AFDC had an average letter-word identification score (`lwss97`) that was 3.17 points lower than that of children who never used AFDC ($p < .05$). Furthermore, the study adjusted selection based on six observed variables (i.e., the ratio of family income to poverty threshold in 1996, caregiver's education in 1997, caregiver's history of using welfare, child race, child age in 1997, and child gender). We now examine the same data in an attempt to determine to what extent is this study sensitive to hidden selection bias.

Results of the sensitivity analysis are shown below. Using the Wilcoxon's signed-rank test, the sensitivity analysis shows that the study becomes sensitive to hidden bias at $\Gamma = 1.43$. Because 1.43 is a small value, we can conclude that the study is very sensitive to hidden bias, and therefore, further analysis that controls for additional biases is warranted.

```r
# Import data and pivot data from long to wide
d <- haven::read_dta(here("data", "adv", "chpt5_2.dta")) %>%
  select(pm, kuse, lwss97) %>%
  drop_na() %>%
  filter(pm != "NA") %>%
  arrange(pm, kuse) %>%
  group_by(pm) %>%
  pivot_wider(names_from = kuse, values_from = lwss97, names_prefix = "y") %>%
  ungroup()
head(d)
```

```
## # A tibble: 6 x 3
##   pm       y0    y1
##   <chr> <dbl> <dbl>
## 1 m.1      81    81
## 2 m.10     97    80
## 3 m.100   131   108
## 4 m.101   105   124
## 5 m.102   141    96
## 6 m.103   102    95
```

```r
p1 <- rbounds::psens(d$y0, d$y1, GammaInc = 0.01)
p1$bounds %>%
  filter(Gamma %in% c(1, 1.3, 1.42, 1.43, 1.5, 2))
```

```
##   Gamma Lower bound Upper bound
## 1  1.00           0      0.0000
## 2  1.30           0      0.0094
## 3  1.42           0      0.0407
## 4  1.43           0      0.0451
## 5  1.50           0      0.0860
## 6  2.00           0      0.7254
```

We use the Wilcoxon signed rank test to test the significance of y. The results show that the randomization test assuming no hidden bias is highly significant ($z = -4.191$, $p < .001$). That is, there is a strong treatment effect.

```r
shapiro.test(d$y1) # non-normal, therefore Wilcoxon signed rank test
```

```
##
##  Shapiro-Wilk normality test
##
## data:  d$y1
## W = 0.98058, p-value = 0.0008692
```

```r
coin::wilcoxsign_test(y1 ~ y0, data = d)
```

```
##
##  Asymptotic Wilcoxon-Pratt Signed-Rank Test
##
## data:  y by x (pos, neg)
##   stratified by block
## Z = -4.1914, p-value = 0.00002772
## alternative hypothesis: true mu is not equal to 0
```

We can further compare the level of sensitivity to bias of the AFDC study to that of other studies. The table below presents results of sensitivity analyses for four observational studies (Rosenbaum, 2005, table 4). Among

Treatment	Gamma = 1	Gamma, max p-value
Study 1: Smoking/Lung Cancer	< .001	(5, 0.03)
Study 2: Diethylstilbestrol/Vaginal Cancer	< .001	(7, 0.054)
Study 3: Lead/Blood Lead	< .001	(4.25, 0.047)
Study 4: Coffee/Myocardial Infarction	.0038	(1.3, 0.056)

the four studies, Study 2 is the least sensitive to hidden bias; the study of the effects of diethylstilbestrol becomes sensitive at about $\Gamma = 7$. In contrast, Study 4 is the most sensitive to hidden bias; the study of the effects of coffee becomes sensitive at about $\Gamma = 1.3$.

Rosenbaum (2005) provided the following explanation of the implications of the sensitivity analyses for these studies: "A small bias could explain away the effects of coffee, but only an enormous bias could explain away the effects of diethylstilbestrol. The lead exposure study, although quite insensitive to hidden bias, is about halfway between these two other studies, and is slightly more sensitive to hidden bias than the study of the effects of smoking" (p. 1812).

Compared with these four studies, our study of the effect of welfare use on academic achievement is slightly better (i.e., less sensitive to unobserved bias) than the study of the effects of coffee, but it is more sensitive to hidden bias than the three other studies.

Hands-On Exercise: Optimal Matching and Sensitivity Analysis

This hands-on exercise uses a simulated dataset (`data.main`) to see if smoking (`tr_smoke`) affects cardiovascular disease (`y_CVD`), with age (`x_age`) as a potentially confounding covariate.[10]

How would you measure the odds of hidden bias, after using 1:1 optimal pair matching without replacement?

```
# Simulated dataset from Zhao et al. (2021)
set.seed(2020)
x_gender <- rep(0:1, c(400, 600))
x_age <- round(abs(rnorm(1000, mean = 45, sd = 15)))
z <- (x_age - 45) / 15 - (x_age - 45)^2 / 100 + 2 * x_gender
tps <- exp(z) / (1 + exp(z))
tr_smoke <- as.numeric(runif(1000) < tps)
z.y <- x_gender + 0.3 * x_age + 5 * tr_smoke - 20
y <- exp(z.y) / (1 + exp(z.y))
y_CVD <- as.numeric(runif(1000) < y)
data.raw <- tibble(x_age, x_gender, tr_smoke, y_CVD)
sub <- sample(1:nrow(data.raw), round(nrow(data.raw) * 8 / 10))
(data.main <- data.raw[sub, c("x_age", "tr_smoke", "y_CVD")])
```

```
## # A tibble: 800 x 3
##    x_age tr_smoke y_CVD
##    <dbl>    <dbl> <dbl>
##  1    56        0     0
##  2    45        1     1
##  3    25        0     0
##  4    33        0     0
##  5    15        0     0
##  6    69        0     1
##  7    55        1     1
##  8    48        1     0
##  9    53        0     0
```

[10] Zhao, Q.-Y., Luo, J.-C., Su, Y., Zhang, Y.-J., Tu, G.-W., & Luo, Z. (2021). *Propensity score matching with R: Conventional methods and new features. Annals of Translational Medicine*, *9*(9), 812. https://doi.org/10.21037/atm-20-3998

```
## 10    24        0     0
## # i 790 more rows
```

```r
# 1:1 Optimal Matching
m.out <- MatchIt::matchit(
  formula =
  data = data.main,
  distance = "logit",
  method = "optimal",
  replace = FALSE,
  ratio = 1
)

# Obtain matched pairs
m.pairs <-

# Count the first set of discrepant pairs in a table of treated and control outcomes
x <-

# Count the second set of discrepant pairs in a table of treated and control outcomes
y <-

# Sensitivity test for matched binary outcomes
rbounds::binarysens(

)
```

Interpretation:

Additional R Packages for Sensitivity Analyses

EValue

"The E-value is defined as the minimum strength of association on the risk ratio scale that an unmeasured confounder would need to have with both the exposure and the outcome, conditional on the measured covariates, to fully explain away a specific exposure–outcome association. As discussed below, the E-value makes no assumptions on whether the unmeasured confounders are binary, continuous, or categorical, on how they are distributed, or on the number of confounders, and it can be applied to several common outcome types in observational research."[11]

For example, "Hammond and Horn estimated that cigarette smoking increased the risk of lung cancer by more than 10-fold. Fisher proposed that genetic confounding may explain all of the apparent association. We can calculate the magnitude of confounding that would be necessary to fully explain the estimated risk ratio of 10.73 (95% CI 8.02, 14.36). The E-value of 20.95 tells us that a confounder, or a set of confounders, such as that proposed by Fisher, would have to be associated with a 20-fold increase in the risk of lung cancer, and must be 20 times more prevalent in smokers than non-smokers, to explain the observed risk ratio. If the strength of one of these relationships were weaker, the other would have to be stronger for the causal effect of smoking on lung cancer to be truly null."[12]

```r
pacman::p_load(EValue)
EValue::evalues.RR(est = 10.73, lo = 8.02, hi = 14.36)
```

```
##             point    lower upper
## RR       10.73000  8.02000 14.36
```

[11] Mathur, M. B., Ding, P., Riddell, C. A., & VanderWeele, T. J. (2018). Website and R Package for Computing E-Values. *Epidemiology*, *29*(5), e45–e47. https://doi.org/10.1097/EDE.0000000000000864

[12] https://cran.r-project.org/web/packages/EValue/vignettes/unmeasured-confounding.html

```
## E-values 20.94777 15.52336    NA
```

konfound

`konfound` "calculates the impact of an omitted confounding variable necessary to invalidate or sustain an inference for a regression coefficient from the user's model. It also assesses how strongly an omitted variable has to be correlated with the outcome and the predictor of interest to invalidate or sustain the inference."[13]

```r
pacman::p_load(konfound)
m1 <- lm(mpg ~ wt + hp, data = mtcars)
tidy(m1)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)  37.2      1.60        23.3  2.57e-20
## 2 wt           -3.88     0.633       -6.13 1.12e- 6
## 3 hp           -0.0318   0.00903     -3.52 1.45e- 3
```

```r
konfound::konfound(m1, wt)
```

```
## Robustness of Inference to Replacement (RIR):
## To invalidate an inference,  66.629 % of the estimate would have to be due to bias.
## This is based on a threshold of -1.294 for statistical significance (alpha = 0.05).
##
## To invalidate an inference,  21  observations would have to be replaced with cases
## for which the effect is 0 (RIR = 21).
##
## See Frank et al. (2013) for a description of the method.
##
## Citation: Frank, K.A., Maroulis, S., Duong, M., and Kelcey, B. (2013).
## What would it take to change an inference?
## Using Rubin's causal model to interpret the robustness of causal inferences.
## Education, Evaluation and Policy Analysis, 35 437-460.

## For more detailed output, consider setting `to_return` to table

## To consider other predictors of interest, consider setting `test_all` to TRUE.
```

[13] Xu, R., Frank, K. A., Maroulis, S. J., & Rosenberg, J. M. (2019). konfound: Command to quantify robustness of causal inferences. *The Stata Journal*, *19*(3), 523–550. https://doi.org/10.1177/1536867X19874223